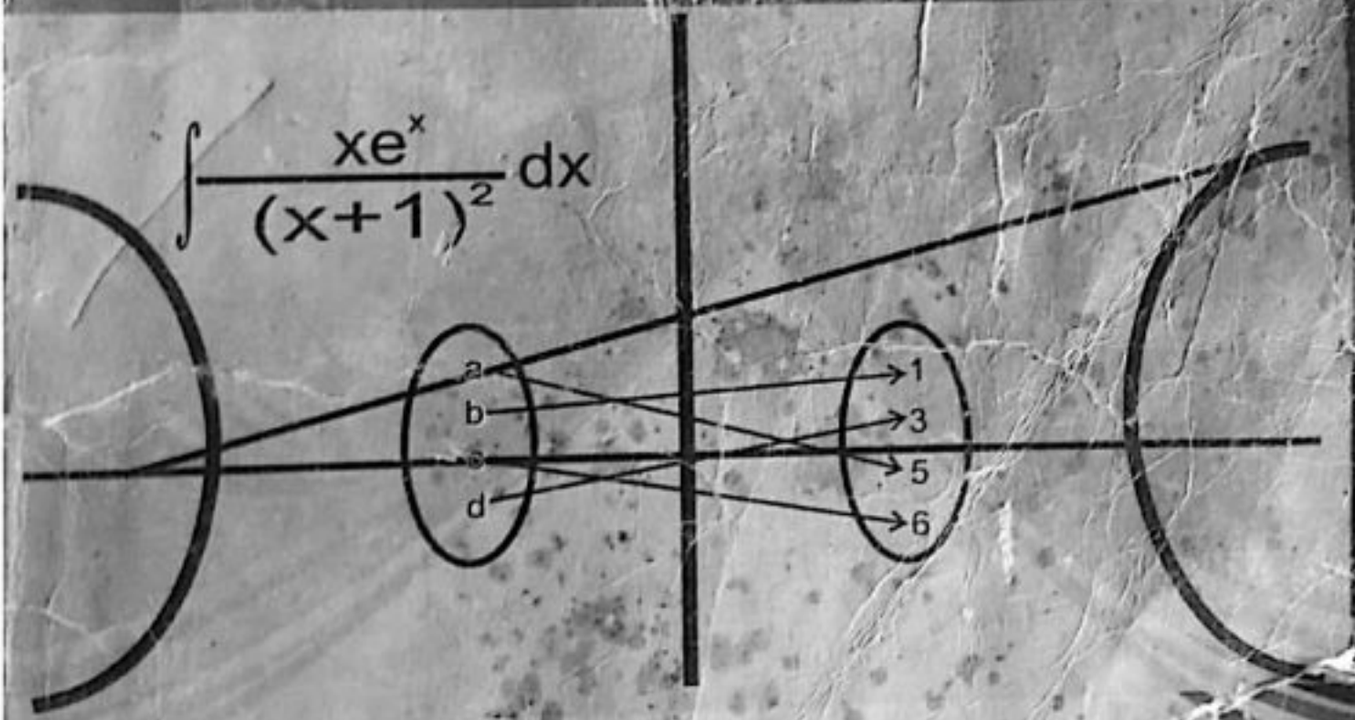
MR. OHMS OUTLINE SERIES

EASY APPROACH IN MAT 102



- Numerous past questions completely solved
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- Both circles & line co-ordinate geometry
- Questions on Conic Sections

1 must for all year one students

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Introduction to Mat 102: Elementary Mathematics.

COURSE OUTLINE:

1. **Concept of functions**
* Functional Notations
*. **Terms Of Functions**
I. Domain ii. Codomain iii. Image of a function
iv. Range
*. Types Of Functions
i. One - One Function (Injective)
ii. Onto Function (Surjective) iii. Bijective function
iv. Solved Examples
*. Constant Function *. Step Function
2. **Concept Of Limit**
3. **Differentiation**
i. Sum Of Differentiation ii. Product Of Differentiation
iii. Quotient Rule iv. Function Of Functions
v. Derivative Of Trigonometry vi. Differentiation Of Exponential Functions
vii. Differentiation Of Logarithms Functions
viii. Differentiation Of General Exponential Functions
ix. Higher Derivatives x. Implicit Functions
*. Applications of Differentiation
i. Tangent to a curve ii. Normal to a curve
iii. Velocity and acceleration iv. Maximum and Minimum Point v. Point Of Inflexion
4. **Integration**
*. Integration Of Exponential Function
*. Integration Of Exponential Function
*. Integration Of Trigonometric Functions
* Integration by part * Definite/Indefinite Integral
* Applications Of Integration

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i. Area Under a Curve ii. Volume Of Revolution

5. Coordinate Geometry

* Distance or Length between two Points
*. Mid Point Of a Straight Line * Division Of a line segment into a given Ratio.
* Gradient Or Slope Of a Straight line
* Equation Of a Straight line

6. Circle

* Equation Of a circle

7. Conic Sections

* Parabola *. Parametric Equation Of a Parabola
* Ellipse * Hyperbola *. Solved Examples.

Concept Of Functions:

By definition, a function is a relation between any two variables such as x and y. In other words, the value of one is dependent upon the value of the other. E.g. If $y = 2x + 1$, then y is a function of x. If $x = 1$,

$y = 2(1) + 1 = 3$. Also, if $x = 2$, $y = 5$. Here, the variable 'x' is called the independent variable while the variable 'y' is called the dependent variable.

Functional Notations:

Generally, a function of x can be denoted by the notation f(x). However, other notations such g(x), h(x), Q(x) and U(x) can also be used to denote functions of x.

Example 1:

If $f(x) = x+3$, find (i) $f(1)$, (ii) $f(0)$, (iii) $f(-4)$, (iv) $f(\frac{1}{2})$

Working: Given $f(x) = x+3$

i. $f(1) = 1 + 3 = 4$ ii. $f(0) = 0 + 3 = 3$

iii. $f(-4) = -4 + 3 = -1$ iv. $f(\frac{1}{2}) = \frac{1}{2} + 3 = 3\frac{1}{2}$

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Example 2:

If $f(x)=2x^2-3x+1$, find:

(i) $f(2)$ (ii) $f(-3)$ (iii) $f(0)$ (iv) $f(½)$

Working: Given $f(x)=2x^2-3x+1$

i. $f(2)=2(2)^2-3(2)+1=3$ ii. $f(-3)=2(-3)^2-3(-3)+1=28$

iii. $f(0)=2(0)^2-3(0)+1=1$ iv. $f(½)=2(½)^2-3(½)+1=0$

Example 3:

If $f(x)=(x+1)/x$ $(x\neq 0)$ and $h(x)=x/(1+x)$ $(x\neq -1)$. Determine

(i). $f(2)+h(½)$ (ii). $\dfrac{f(2)-h(½)}{2}$

Working: (i) Given, $f(2)+h(½)$,

Where $f(x)=(x+1)/2$, $\Rightarrow f(2)=(2+1)/2=3/2$; $h(x)=x/(1+x)$, $\Rightarrow$

$h(½)=\dfrac{½}{1+½}=1/3$. therefore, $f(2)+h(½)=\dfrac{3}{2}+\dfrac{1}{3}=\dfrac{11}{6}$

(ii). $\dfrac{f(2)-h(½)}{2}=\dfrac{3/2-1/3}{2}=\dfrac{7}{12}$

Example 4: Let $g(x)=x^2$ and $f(x)=x+1$, find i. $f(3)/g(2)$

(ii). $gof(x)$ (iii). $fog(x)$ iv. $gof(½)$ v. $gof(2)+fog(0)$

Working:

Given $f(x)=x+1$ and $g(x)=x^2$. ***therefore,*** $f(3)=3+1$ and

$g(2)=2^2=4$. *THUS,* (I). $f(3)/g(2)=4/4=1$

ii). $gof(x)=g(f(x))=g(x+1)=(x+1)^2$

iii). $fog(x)=f(g(x))=f(x^2)=x^2+1$

iv). $gof(2)=(2+1)^2=9$ (v). $gof(2)+fog(0)=9+1=10$

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Example 5:

Functions of f and g are defined by $f: x \longrightarrow 3x-4$ and $g: x \longrightarrow -2x+3 (x \in R)$.

Find; (i) $f^{-1}(x)$ (ii) $g^{-1}(x)$ (iii) fog (iv) gof (v) fog^{-1}

Working:

Given $f(x)=3x-4$, $g(x)=-2x+3$

1. $f^{-1}(x)$ = Inverse of $f(x)$; let $f(x)=y$

   $\Rightarrow y=3x-4$ .i.e $y+4=3x$; $x = \frac{y+4}{3}$; $\therefore f^{-1}(x) = \frac{x+4}{3}$

ii. $g^{-1}(x)$ = inverse of $g(x)$, let $f(x)=y$

   $\Rightarrow g(x)=y=-2x+3$; $2x=3-y$; $x=\frac{3-y}{2}$. $\therefore g^{-1}(x) = \frac{3-x}{2}$

iii. $fog(x)=f(g(x))=f(-2x+3)=3(-2x+3)-4 = -6x+5$

iv. $gof=g(f(x))=g(3x-4)=-2(3x-4)+3=-6x+11$

v. fog^{-1} = inverse of fog. Recall that $fog=-6x+5$

   $\Rightarrow fog=y=-6x+5$; $6x=5-y$; $x=\frac{5-y}{6}$, $\therefore fog^{-1} = \frac{5-x}{6}$

Terms Of Functions:

This include;

i. ***Domain:*** Domain is defined as all values of 'X' that will not make the given function Undefined. Simply put, the set X (the departure set) is called the domain of F.

ii. ***Co-domain*** : The set Y (the arrival set) is called the Co-domain of F.

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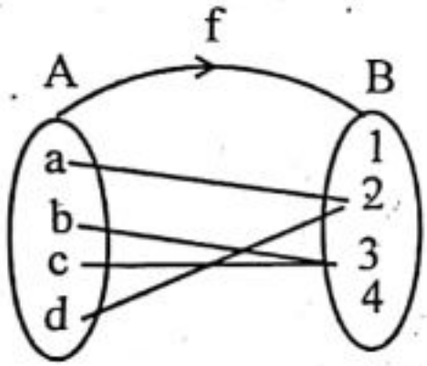
NB; Departure set and arrival sets are also referred to as "*object set* and "*image set*" respectively.

iii. ***Image Of An Element:*** For each element $x \in X$, the element $y \in Y$ such that $f(x) = y$ is called ***image*** of x.

iv. **Range:** Let $f: X \to Y$; The range of f is the set of those elements in Y which appear as the image of at least one element in X. Also, the range of a map (function) is defined to consist precisely of those elements in the Co-domain which appear as the image of at least one element in the domain.

Example: let $f: A \longrightarrow B$

i.e.



From the above,

(i). Domain of A = {a, b, c, d} (ii) Codomain = {1,2,3,4}

(iii) Image of A = {2,3} (iv) Range = {2,3}

NB;. (1) The range of f is synonymous with image of the set.

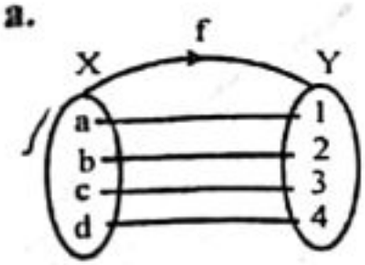
i.e. f(A)

Condition For A Relation to Be A Function:

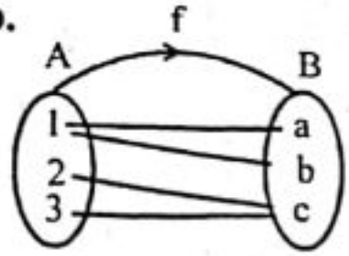
For a relation to satisfy the requirement for a function, every element in the *domain* must correspond to "one" and only "one" element in the *codomain.* i.e

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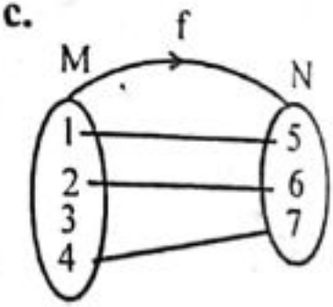
a.



b.



c.



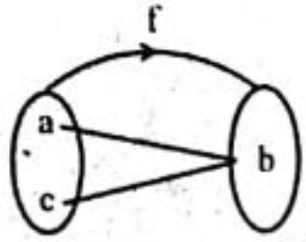
from the figure above, only option (a.) defines a function.

Example 2: Which of the following relation is a function?

(a). $(a, a), (a, b)$ b. $(a, b), (c, b)$ c. $(b, d), (b, -d)$

(d). None.

Working:



THUS, the correct option is B

Example 3: State which sets of the following ordered pairs is a function. (a). $\{(x,y)/y=x+4\}$ b. $(x,y)/x=6$ c. $\{(x,y)/x+y>4\}$ d. $\{(x,y)/y>x+1\}$ e. $\{(x,y)/x+2=0\}$

Working:

The correct option is A. This is so because;

(i). The function ($y=x+4$) contains x and y in the equation.

(ii). The function contains only the '' equality sign (=)" and does not contains either '<' or '>' sign.

Example 4: Only one of the following sets of ordered pairs is a function. a. $\{(x,y)/y>x+1\}$ b. $(x,y)/y+2=0$ c. $\{(x,y)/x=6\}$ d. $(x,y)/x^2+y^2<25$

Working: From the options above, it was only option B that

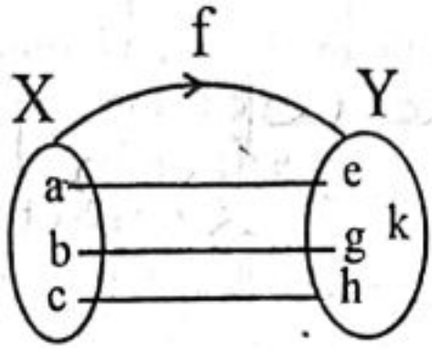
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defines a function. ***thus***, option is **B** is correct.

Types Of Function/Mapping:

i. **One-One Function (Injective):** A function f: X ⟶ Y is said to be one-one, if every pair of different element in X have different image in Y. Equivalently, a function f: X ⟶ Y is one - one if $f(x)=f(a')$.

Illustration 1:



Also, a function is said to be one-one if no two elements in the domain have the same image in the **CO-DOMAIN.**

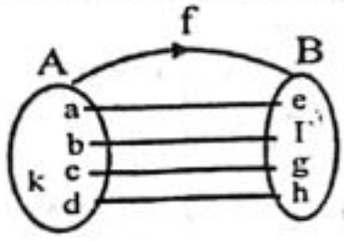
Illustration 2:

The map f: R ⟶ R defined by f: x ⟶ x-2 on the set of real numbers is one - one. Verify that f is one- one. (Get Serious)

(2). Onto Functions (surjective):

f: X ⟶ Y is said to be onto if every element of Y is the image of at least one element of X. We then say that f maps X onto Y. Equivalently, a function f from X to Y is onto, if its range is equal to its *CODOMAIN*. i.e. f(x) = Y.

Illustration 1:



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The above defines an ***Onto function***.

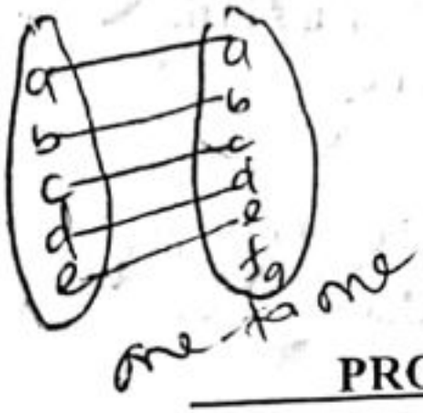
Illustration 2: Let f: X → Y, where X = {states in Nigeria}
Y = {capital of states in Nig} Then, f is both one-one and onto.
Domain = {state in Nigeria}, Codomain = {capital of st. in Nig}
Range = {capital of states in Nigeria}

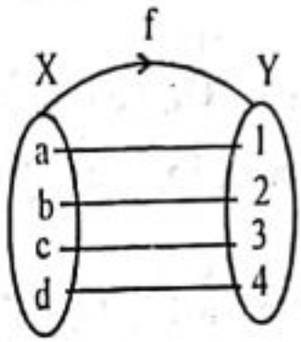
3 **Bijective Function:**

A function which is both one-one and onto is said to be *BIJECTIVE.* e.g the ~~function~~ relation of Nig. State & Capital)

4 **One-one Correspondence:**

If a function is both one-one and onto, it is said to define a one-one correspondence i.e.





PROBLEMS:

Question 1: Let A = {-3,2,1}, Z = Sets of integers, let f: A → Z be defined by f(x) = 2x - 1. Find i. Domain of f
ii. Codomain of f iii. Range of f iv. Image of -3
v. Is f one-one vi. Is f onto

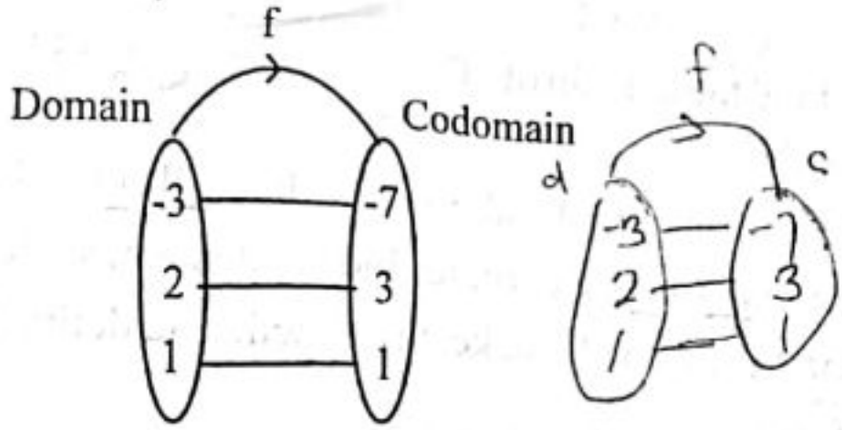
Working:

Given A = {-3,2,1} f(x) = 2x - 1 i. Domain of f = {-3,2,1}
ii. Codomain = {-7,3,1}. i.e. If f(x) = 2x - 1, when x = -3
=> f(-3) = 2(-3) -1 = -7. When x = 2,

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$= f(2) = 2(2) - 1 = 3$. When $x = 1$, $\Rightarrow f(1) = 2(1) - 1 = 1$

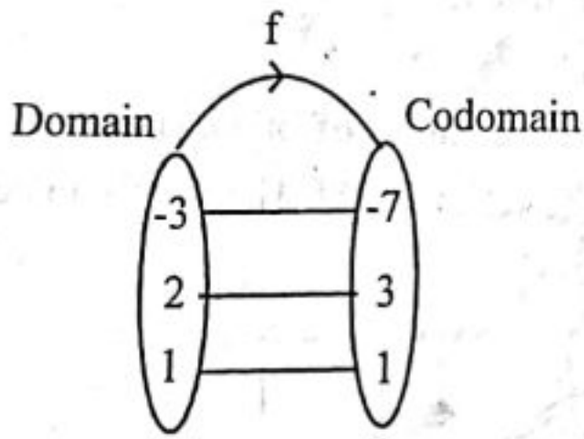
thus, we have;



(vii) Range of $f = \{-7, 3, 1\}$ = Sets of all integers

(v) Image of $(-3) = 2(-3) - 1 = -7$

(vi) Using the diagram below; i.e.



From the diagram above, F is One-one. This is so because, every pair of different elements in the ***DOMAIN*** have different images in the ***CODOMAIN.***

(vi). f is Onto, since every member of the ***CODOMAIN*** is the image of at least one member in the ***DOMAIN***

Question 2:

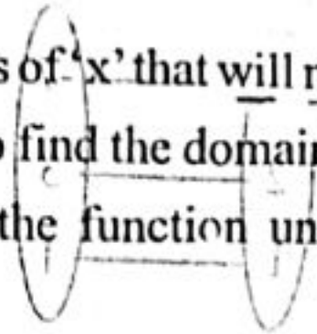
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Let $f(x) = \frac{1}{x+4}$ $x \in R$

(i). Find the domain of f (ii). The range of f

Working:

Domain is defined as all values of 'x' that will not make the given function ***undefined***. Hence, to find the domain, we look for that value of 'x' that will make the function undefined and then ***exclude*** it.



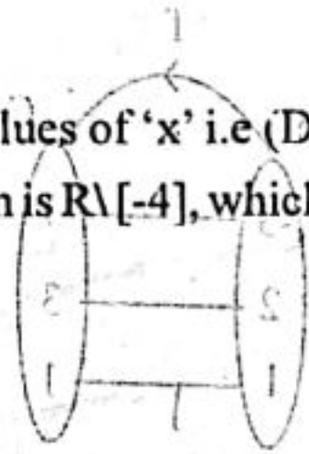
Let's go

For the above function to be undefine, the denominator must be equal to zero. i.e $x+4=0$, ***thus***, $x=-4$

Since putting $x=-4$ in the above function we will have;

$$\frac{1}{-4+4} = \frac{1}{0} = \textit{undefined},$$

We then exclude -4 from the values of 'x' i.e (Domain). ***Hence***, the domain of the above function is $R\backslash[-4]$, which can be read as 'all real numbers except -4'



(ii). Range

Step 1:

To obtain the range, re-write the given function as $y = \frac{1}{x+4}$

Step 2: Make x the subject

i.e $y(x+4)=1$; $x+4=\frac{1}{y}$; $x=\frac{1}{y}-4$

Step 3: Find the value that will make the function ***undefined***

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and *exclude* it.

Now, for the $x = \frac{1}{y} - 4$ to be undefined, y must be equal to zero. i.e

$x = \frac{1}{0} - 4 = \varphi$ = undefined.

Hence, the Range is R\[0], i.e all real numbers except 0

Question 3:

Define f: R R by $f(x) = \frac{x^2}{x^3+x^2-4x-4}$

(i). For what values of x is f undefined

(ii). Hence or otherwise determine the domain of f.

Working:

(i). For f to be undefined, the denominator must be equal to zero.

i.e $x^3+x^2-4x-4=0$

By guess, put $x=-1$

$(-1)^3+(-1)^2-4(-1)-4=0$; $-1+1+4-4=0$

Divide the function by $x+1$ to get other functions

$$
\begin{array}{r|l}
 & x^2 - 4 \\
\hline
x+1 & x^3+x^2-4x-4 \\
 & -\,x^3+x^2 \\
\hline
 & -4x-4 \\
 & -\,-4x-4 \\
\hline
 & 0
\end{array}
$$

$x^2 - 4$ is the factor which is the same as $x^2 - 2^2$, ***Hence***, the two other factors are (x-2)(x+2) from difference of two squares.

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$\therefore x^3+x^2-4x-4=(x+1)(x-2)(x+2)$

The values of x for which f will be undefined are

$(x+1)=0,\ x=-1$ $x+1=0, x=-1$

$(x-2)=0,\ x=2$ $x-2=0, x=+2$

$(x+2)=0, x=-2$ $x+2=0, x=-2$

(ii). Domain will be all real numbers except these three numbers above. .i.e -1, 2 & -2, which can be written as R\[-2, -1, 2]

Question 4: All real nos except -2, -1, 2

Given the function, $f(x) = \frac{x}{3-x}$ from R to R, $3-x\neq 0$ $3\neq x$

determine the domain and the range.

Working:

Domain as usual will be to exclude the values of x that will make the function undefined. ***thus***, Domain = R\[3]

To find the range, make x the subject.

$y = \frac{x}{3-x}$; $(3-x)y=x$; $3y=x+xy$; $x=\frac{3y}{1+y}$ $(3-x)y=x$; $3y-xy=x$

thus, Range = R\[-1]

Question 5:

Let f(x) = tanx. Determine

(i). The domain of f (ii). The range of f

(iii). Is f one - one (iv). Is f onto

Working:

$f(x)=\tan x=\frac{\sin x}{\cos x}$. f will be undefined if $\cos x=0$

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or $x = \cos^{-1} 0 = \pi/2$.

thus, Domain $= R = \{x: \cos x = 0\}$ or $R - \left\{\frac{\pi + m\pi}{2}; m \in Z\right\}$

We have $m\pi$. Since the periodic function of $\tan\theta = \pi$,

Range $= y = \tan x$, $\tan^{-1} y = x$ Note that for any value you put in for y, x will have a value. ***Hence***, the Range = R. i.e Sets of all real numbers.

(iii). f is not ***one to one*** function, since two values in the ***Domain*** can give one value in the ***Codomain***. e.g $\tan 30 = 0.57735$ and $\tan 210 = 0.57735$.

(iv). f is ***onto***, since every member of the ***Codomain*** is the image of at least one member of the ***Domain.***

Question 6:

Define $g: R \longrightarrow R$, $g(x-2) = 4x^2 + x + 7$, compute

(i). $g(0)$ $g(-3)$

Working:

We consider a case where $g(x-2)$ is now $= g(0)$.

i.e $g(x-2) = g(0)$

$x - 2 = 0$

$x = 2$, this means at $g(0)$, $x = 2$

Putting $x = 2$ into $4x^2 + x + 7$, we have $g(0) = 4(2)^2 + 2 + 7 = 25$

Similarly, $g(-3) = g(x-2)$

$-3 = x - 2$

$x = -3 + 2$

$x = -1$

$g(-3) = 4(-1)^2 + 1 + 7 = 12$

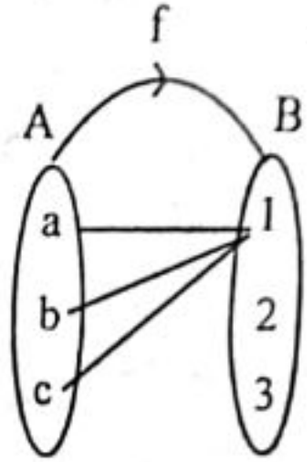
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Constant Function:

A mapping f of A into B is called a ***constant mapping*** if the same element is the image of every element in A. in other words, $f: A \to B$ is a constant mapping if the range of f consists of only one element, that is $f(A) = b \in B$

Example

Let the mapping f be defined by :



then, f is a ***constant mapping*** since '1' is the image to every element in the ***DOMAIN***.

Step Function:

A functions f(x) is called *step function* if it assumes a distinct constant value in each interval of the ***Domain***.

Question 1:

Let a step function be defined by the relation

$$f(x) = \begin{cases} -3 & \text{for } x \le -6 \\ -1 & \text{for } -6 < x \le 2 \\ 4 & \text{for } x > 2 \end{cases}$$

Find the value of: (i) f(-10) (ii) f(20) (iii) f(0) (iv) f(6000)

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~~(i=1,0,...-6 0 1 2 3~~
-5, -4, -3, -2, -1, 0, 1, 2

Working:

$x \leq -6 = \{-\infty, \text{---} 8, -7, -6\}$

$-6 < x \leq 2 = \{-5, -4, -3, -2, -1, 0, 1, 2\}$

$x > 2 = \{3, 4, \text{----} +\infty\}$. ***thus,***

(i). $f(-10) = -3$, since $x = -10$ falls within the range of $x \leq -6$

ii) $f(20) = 4$, since $x = 20$ falls within the range of $x > 2$

iii). $f(0) = -1$, since $x = 0$ falls within the range of $-6 < x \leq 2$.

i. $f(6000) = 4$

Question 2:

Let $f: R \longrightarrow R$ be defined by

$$f(x) = \begin{cases} 3x - 2 \text{ for } x \geq 4 \\ x^2 + 2 \text{ for } -2 \leq x < 4 \\ 2x + 4 \text{ for } x < -2 \end{cases}$$

i). Draw the graph of this function

ii). What is the domain of f? (iii). What is the range of f?

iv). Is f one - one? (v). Is f onto (vi). Find f(-2)

Working:

i). To draw the graph, we consider 2 point from each interval.

For $x \geq 4$, take f(5) & f(4)

$f(5) = 3(5) - 2 = 13$; $f(4) = 3(4) - 2 = 10$

For $-2 \leq x < 6$, take f(-2), f(2), f(0)

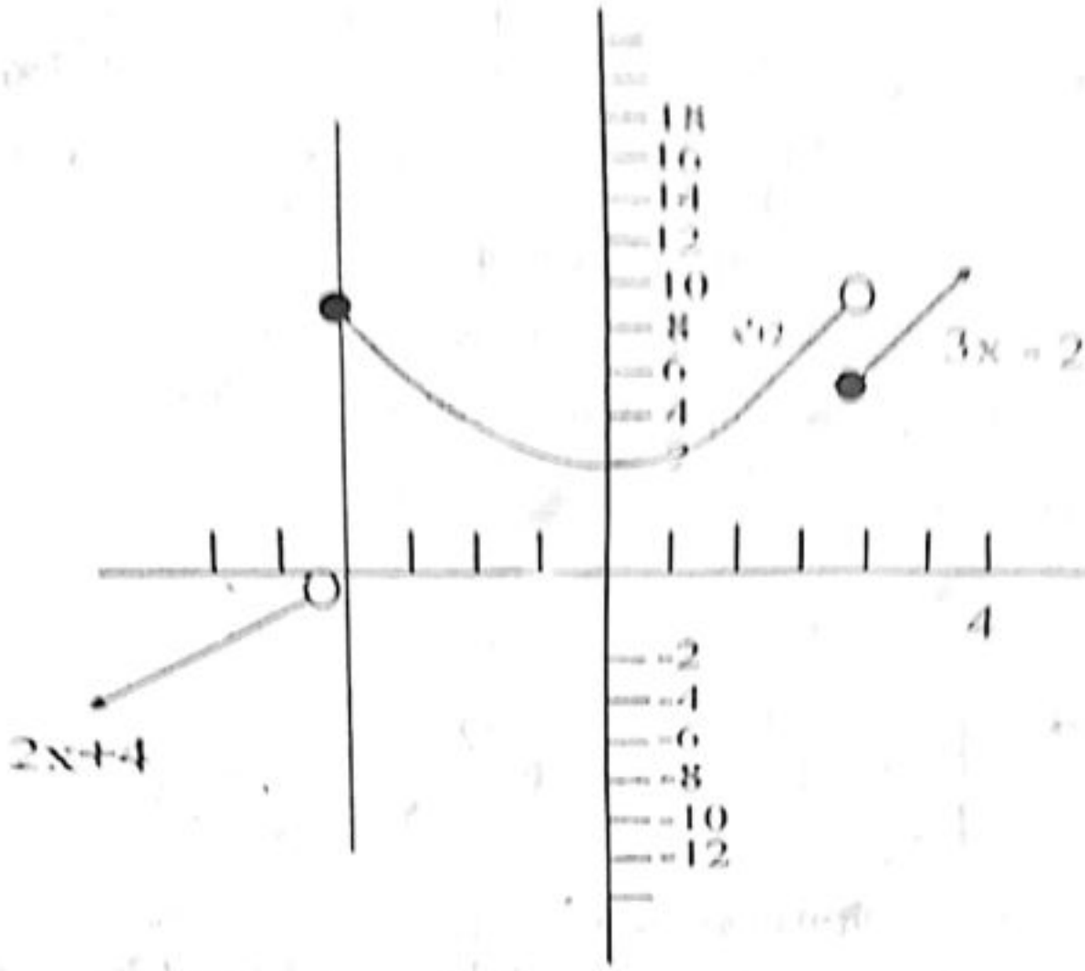
$f(-2) = -2^2 + 2 = 6$; $f(0) = 0^2 + 2 = 2$; $f(2) = 2^2 + 2 = 6$

For $x < -2$, take f(-3) & f(-6)

$f(-3) = 2(-3) + 4 = -2$; $f(-6) = 2(-6) + 4 = -8$

Now, use the above points to plot the graph.

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Explanation of the graph;

1. 3 different graphs were plotted, 2 linear graphs i.e 3x - 2 and 2x + 4, plus one quadratic $x^2 + 2$
2. Each of these graphs were restricted on where to reach and where not to reach on x - axis.
3. So, all we need do is to get two or more point (as in the case of the quadratic graph) but within the restricted values of x to plot each of the individual graphs as stated above.

(ii). Domain = All real numbers. i.e all possible values of x in

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the x-axis

(iii). Range of f = R i.e sets of all real numbers .i.e the uppermost to the lowest value in the y-axis of the given graphs

(iv). f is not one - one, since there are point in the domain such that each points in the domain, for instance, 6 in the codomain is the image of 2 and -2.

(v). f is onto since every member of the codomain is the image of at least one member of the domain

(vi). Find f(-2)

This falls within $-2 \leq x < 4$, whose $f(x) = x^2 + 2$,

this implies that $f(-2) = (-2)^2 + 2 = 6$

Exercise:

Find the Domain and the range of the followjng functions

(i). $f(x) = \sin x + 2\cos x$ (ii). $f(x) = \dfrac{x^2+4}{x^2+1}$

Answers:

(i). Domain = All real numbers, Range = $[-2, 2]$

(ii). Domain = All real numbers, Range = $[1, 4]$

Concept of Limits:

Question 1:

Find Lt. $x^2 + 2x + 1$

$x \to 2$

Working:

First, we substitute for $x = 2$. ***thus,*** $x^2 + 2x + 1 = 2^2 + 2(2) + 1 = 9$

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therefore, $\lim_{x \to 2} x^2+2x+1=9$

Question 2:

Limit $(3x-4)(3x^2+8x-3)$

$x \to -3$

Working:

First, put $x=-3$. So, $[3(-3)-4)(3(-3)^2+8(-3)-3]=0$

therefore, $\lim_{x \to -3} (3x-4)(3x^2+8x-3)=0$

Question 3:

Evaluate the limit

Limit $\dfrac{x-1}{2x^2-x-1}$

$x \to 1$

Working:

Put $x=1$, i.e. $\dfrac{1-1}{2(1)^2-1-1} = \dfrac{0}{0}$ = indeterminate.

So, we apply L' hospital rule by differentiating both the numerator and denominator i.e. $\dfrac{x-1}{2x^2-x-1} = \dfrac{1}{4x-1}$

So, we let $x=1$, i.e $= \dfrac{1}{4(1)-1} = \dfrac{1}{3}$

NB:

Whenever the limit of $x \to a$, (where $a=0, 1, 2, <\infty$) we substitute first into the function.

Question 4:

Evaluate the following;

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(i). $\lim_{x\to\infty} \dfrac{4x^2 + x - 1}{3x^2 - 2x + 1}$

(ii). $\lim_{x\to\infty} \dfrac{x - \sin x}{x^{3/2}}$

(iii). $\lim_{x\to\infty} \dfrac{1 - x}{\cos x}$

Working:

Step 1: We will substitute the values of ∞ into each of the given functions.

So let's go

(i). $\dfrac{4(\infty)^2 + \infty - 1}{3(\infty)^2 - 2(\infty) + 1} = \dfrac{\infty}{\infty}$.

Notice that $\dfrac{\infty}{\infty}$ is an indeterminate value.

Hence, direct substitution at ∞ into the given function will not be appropriate. therefore, we use one of the following methods:

(i). Le - hopital's Rule

(ii). Factorization method (i.e if it is factorizable)

(iii). Thumb's rule . i.e dividing by the highest power of the independent variable.

Note: other indeterminate forms include:

(i). $\dfrac{\infty}{\infty}$ i.e $\infty \div \infty$

(ii). $\infty - \infty$ (iii). $0 \times \infty$ (iv). ∞^0 (v). 1^∞

(vi). $\dfrac{0}{0}$ .i.e $0 \div 0$

(vii). 0^0.

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So students should ''watch out" for the above seven indeterminate forms. Once direct substitution gives you any of the above indeterminate forms, use any of the 3 methods stated above.

So, let us solve question (i) using Thumb's Rule.

Divide both the numerator and the denominator by the highest power of x and the highest power of x is x^2.

thus, we have

$$\frac{\frac{4x^2}{x^2} + \frac{x}{x^2} - \frac{1}{x^2}}{\frac{3x^2}{x^2} - \frac{2x}{x^2} + \frac{1}{x^2}} = \frac{4 + \frac{1}{x} - \frac{1}{x^2}}{3 - \frac{2}{x} + \frac{1}{x^2}}$$

Then substitute ∞ now in your result

$$\frac{\frac{4}{1} + \frac{1}{\infty} - \frac{1}{\infty^2}}{\frac{3}{1} - \frac{2}{\infty} + \frac{1}{\infty^2}}$$

Where $1/\infty = 0$, $1/\infty^2 = 0$. Hence, we have $\frac{4 + 0 - 0}{3 - 0 + 0} = \frac{4}{3}$

Now, let us use Le - Hopital rule for the same above questio

LE - HOPITAL'S RULE:

You differentiate both the numerator and denominator *until* the substitution of the value of x will ***no longer*** give an indeterminate function.

Note : Le Hopital's rule is used ***majorly*** for indeterminate forms.

Now, let's go

$$\frac{4x^2 + x - 1}{3x^2 - 2x + 1}$$

Differentiate $4x^2 + x - 1 = 8x + 1$

Differentiate $3x^2 - 2x + 1 = 6x - 2$

We have $\frac{8x + 1}{6x - 2}$

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Putting $x = \infty$, we have $\frac{8(\infty)+1}{6(\infty)-2} = \frac{\infty}{\infty}$

Since $\frac{\infty}{\infty}$ is still indeterminate, we will differentiate

$\frac{8x+1}{6x-2}$ again until the substitution of $x = \infty$ will be meaningful or until we obtain a constant.

Hence, Differentiate $8x - 1 = 8$, and differentiate $6x - 2 = 6$

thus, we have $\frac{8}{6} = \frac{4}{3}$ the same as above.

(ii). $\lim_{x \to 0} \frac{x - \sin x}{x^{3/2}}$

Step 1: direct substitution gives $\frac{0 - \sin 0}{0^{3/2}} = \frac{0}{0}$ = Indeterminate

So, we use Le - Hopital rule. i.e. $\frac{x - \sin x}{x^{3/2}} = \frac{1 - \cos x}{3/2x^{1/2}}$

Putting $x = 0$, $\frac{1 - \cos 0}{3/2(0)^{1/2}} = \frac{1-1}{0} = \frac{0}{0}$ = Indeterminate

Hence differentiate again. i.e $\frac{1 - \cos x}{3/2x^{1/2}} = \frac{0 - (-\sin x)}{3/4x^{-1/2}} = \frac{\sin x}{3/4x^{-1/2}}$

$= 4/3x^{1/2}\sin x$

Put $x = 0$, we have $4/3(0)^{1/2}\sin(0) = 0$

therefore, answer = 0

NB: Whenever, the limit of $x \to \infty$, we divide first each of the term by the highest power of x.

Question 5:

Evaluate limit $\lim_{x \to \pi} \frac{\sin x}{x - \pi}$

Working:

Put $x \to \pi = \frac{\sin \pi}{\pi - \pi} = \frac{0}{0}$ = Indeterminate

Next, we apply Le - hospital rule

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i.e $\frac{\sin x}{x-\pi} = \frac{\cos x}{1-0} = \frac{\cos\pi}{1} = \frac{-1}{1} = -1$

Question 6:

Evaluate limit $\lim_{x\to 0} \frac{x^3+2x^2+x}{x^2+3x}$

Working:

$\frac{x^3+2x^2+x}{x^2+3x} = \frac{0^2+2(0)^2+0}{(0)^2+3(0)} = \frac{0}{0}$ = indeterminate

Use Le-hospital rule

i.e $\frac{x^3+2x^2+x}{x^2+3x} = \frac{3x^2+4x+1}{2x+3}$

Putting $x=0$, we have $\frac{3(0)^2+4(0)+1}{2(0)+3} = \frac{1}{3}$

Question 7:

Evaluate (i) Limit $\lim_{x\to 1} \frac{x^2-7x+6}{x^2-x}$ (ii). Lt $\lim_{x\to 0} \frac{\sin 2x-2}{x^3}$

Working:

(i). $\lim_{x\to 1} \frac{x^2-7x+6}{x^2-x} = \frac{1^2-7(1)+6}{1^2-1} = \frac{0}{0} = \infty$

Notice ∞ is ***undefined*** and <u>not</u> ***indeterminate***, so it is answer

(ii). $\frac{\sin 2x-2x}{x^3} = \frac{\sin 2(0)-2(0)}{0^3} = \frac{0}{0}$ = Indeterminate

Use Le-Hospital

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$$\frac{\sin 2x - 2x}{x^3} = \frac{2\cos 2x - 2}{3x^2} = \frac{2\cos 0 - 2}{3(0)^2} = \frac{2-2}{0} = \frac{0}{0} = \text{Indeterminate}$$

Use Le - hospital

i.e $\frac{2\cos 2x - 2}{3x^2} = \frac{-4\sin 2x}{6x} = \frac{-4\sin 0}{6(0)} = \frac{0}{0} = \text{Indeterminate}$

Use Le - hospital again the 3rd time.

$$\frac{-4\sin 2x}{6x} = \frac{-8\cos 2x}{6} = \frac{-8\cos 2(0)}{6} = \frac{-8\cos 0}{6} = \frac{-8}{6} = \frac{-4}{3}$$

Question 8:

Evaluate the following: (i).

$$\lim_{x \to 0} \frac{(1 - \cos x)}{2x^2 - x} = \frac{1 - \cos(0)}{2(0)^2 - 3(0)} = \frac{1-1}{0} = \frac{0}{0} = \text{Indeterminate}$$

Using Le - hopital, $\frac{\sin x}{4x - 1} = \frac{\sin(0)}{4} = \frac{0}{4} = 0$

(ii). $\lim_{x \to 0} \left(\frac{x}{\sin x} + \cot x \right)$

Working:

Since $\cot x = \frac{\cos x}{\sin x}$; $\frac{x}{\sin x} + \cot x = \frac{x}{\sin x} + \frac{\cos x}{\sin x}$

Putting $x = 0$, $= \frac{x + \cos x}{\sin x} = \frac{0 + \cos 0}{\sin(0)} = \frac{0+1}{0} = \infty$

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Question 9:

Evaluate limit $\underset{x\to 0}{} \dfrac{8-\sin x}{2x} = \dfrac{8-\sin 0}{2(0)} = \dfrac{8-0}{0} = \infty$

Note that ∞ is *Undefined* and not *Indeterminate*.

Exercise:

Evaluate the following limits.

1. $\underset{x\to 2}{\text{Lt.}} \dfrac{x^3-4}{x^3-2x}$

2. $\underset{x\to 0}{\text{Lt.}} \dfrac{3x+1}{x}$

3. $\underset{x\to 2}{\text{Lt.}} \sqrt{2x^3-3x+18}$

4. $\underset{x\to -\infty}{\text{Lt.}} \sqrt{\dfrac{x^3-1}{x^4-4x^3-5x^2+3}}$

CONCEPT OF DIFFERENTIATION:

By definition, ***Differentiation*** is the process of finding the differential coefficient of one variable with respect to another, and can be written mathematically as ***thus***:

$$\frac{dy}{dx} = \text{Lt}\ \frac{f(x+\delta x) - f(x)}{\delta x}$$

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Generally, when $y = x^n$, ***then*** $dy/dx = nx^{n-1}$. ***thus,*** if;

i. $y = x^4$, $dy/dx = 4x^3$ (ii). $y = 10x^{-7}$, $dy/dx = -70x^{-8} \equiv -70x^{-1}$

NB:

1. $\frac{dy}{dx} = \frac{1}{dx}\left(dy\right) = y' = f'(x)$

2. $\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = y'' = f''(x)$

3. $\frac{d^3y}{dx^3} = \frac{d}{dx}\left(\frac{d^2y}{dx^2}\right) = y''' = f'''(x)$

Differentiation From first Principles:

Question 1:

Using first principle evaluate dy/dx for $y = 2x^2 + 4x$.

Working:

Step 1: Increase both sides by a small increment

$y + \delta y = 2(x + \delta x)^2 + 4(x + \delta x)$

$y + \delta y = 2(x^2 + 2x\delta x + (\delta x)^2) + 4(x + \delta x)$

Step 2: Subtract y from both sides

$y + \delta y - y = 2(x^2 + 2x\delta x + (\delta x)^2) + 4(x + \delta x) - y$

But $y = 2x^2 + 4x$

$\delta y = 2(x^2 + 2x\delta x + (\delta x)^2) + 4x + 4\delta x - 2x^2 - 4x$

Re-arranging

$\delta y = 2x^2 + 4x\delta x + 2(\delta x)^2 + 4x + 4\delta x - 2x^2 - 4x$

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$\delta y = 4x\delta x + 2(\delta x)^2 + 4\delta x$

Step 3: Divide through by δx

$$\frac{\delta y}{\delta x} = \frac{4x\delta x}{\delta x} + \frac{2(\delta x)^2}{\delta x} + \frac{4\delta x}{\delta x}$$

$$\delta y = 4x + 2(\delta x) + 4$$

$\delta x \to 0$

$$\therefore \frac{dy}{dx} = 4x + 2(0) + 4 = 4x + 4$$

Question 2:

Differentiate from first principle $y = \frac{1}{\sqrt{x}}$

Working:

$$y = \frac{1}{\sqrt{x}}$$

Step 1: Square both sides

$$y^2 = \frac{1}{x}$$

Step 2: Increase both sides by small differential

$$(y + \delta y)^2 = \frac{1}{x + \delta x}$$

Step 3: Subtract y2 from both sides

$$(y + \delta y)^2 - y^2 = \frac{1}{x + \delta x} - y^2$$

But $y^2 = \frac{1}{x}$

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$$(y+\delta y)^2 - y^2 = \frac{1}{x+\delta x} - \frac{1}{x}$$

Expanding, we have $y^2 + 2y\delta y + (\delta y)^2 - y^2 = \frac{1}{x+\delta x} - \frac{1}{x}$

$$\text{i.e } 2y\delta y + (\delta y)^2 = \frac{x-(x+\delta x)}{(x+\delta x)(x)}$$

$$2y\delta y + (\delta y)^2 = \frac{x-x-\delta x}{x(x+\delta x)}$$

$(\delta y)^2 \to 0$ product of small difference

$$2y\delta y = \frac{-\delta x}{x(x+\delta x)}$$

Step 4: Divide both sides by δx

$$\frac{\delta y}{\delta x} = \frac{-\delta x}{2yx(x+\delta x)\,\delta x} = \frac{-1}{2xy(x+\delta x)}$$

$$\text{If } \delta x \to 0,\ \frac{dy}{dx} = \frac{-1}{2xy(x+0)} = \frac{-1}{2yx^2}$$

$$\text{But } y = \frac{1}{\sqrt{x}}$$

$$\textit{therefore},\ \frac{dy}{dx} = \frac{-1}{2x^2\left(\frac{1}{\sqrt{x}}\right)}$$

Question 3:

Differentiate from first principle $y = \frac{1}{x} + 3x^2$

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Working:

Step 1: Increase both sides by a small increment

$$y+\delta y = \frac{1}{(x+\delta x)} + 3(x+\delta x)^2$$

Step 2: Subtract y from both sides

$$y+\delta y - y = \frac{1}{x+\delta x} + 3(x+\delta x)^2 - y \text{ . But } y = \frac{1}{x} + 3x^2$$

$$\delta y = \frac{1}{x+\delta x} + 3(x+\delta x)^2 - \frac{1}{x} + 3x^2$$

Re-arranging

$$\delta y = \frac{1}{x+\delta x} - \frac{1}{x} + 3(x+\delta x)^2 - 3x^2$$

$$\delta y = \frac{1}{x+\delta x} - \frac{1}{x} + \cancel{3x^2} + 6x\delta x + 3(\delta x)^2 - \cancel{3x^2}$$

$$\delta y = \frac{1}{x+\delta x} - \frac{1}{x} + 6x\delta x + 3(\delta x)^2$$

$$\delta y = \frac{x-(x+\delta x)}{x(x+\delta x)} + 6x\delta x + 3(\delta x)^2$$

Step 3: Divide through by δx

$$\frac{\delta y}{\delta x} = \frac{-\cancel{\delta x}}{x(x+\delta x)\cancel{\delta x}} + \frac{6x\cancel{\delta x}}{\cancel{\delta x}} + \frac{3(\delta x)^{\cancel{2}}}{\cancel{\delta x}}$$

$\delta x \to 0$

$$\frac{dy}{dx} = \frac{-1}{x(x+0)} + 6x + 3(0) = \frac{-1}{x^2} + 6x$$

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Exercise:

Find from first principles the differential coefficient of the following functions.

(i). $y = \sqrt{x}$ (ii). $y = x^3$

(iii). $y = 1/2x$ (iv). $y = 1/x$

(v). $y = 4x^2 + 3x$ (vi). $y = x^3 + x^2$

RULES OF DIFFERENTIATION:

1. Sum of differentiation:

Let $y = u + v$

Where; u = function of x, v = function of x

thus, by formula, $dy/dx = du/dx + dv/dx$

Question 1:

$y = 3x^2 + 2x$, find dy/dx

Working:

let $u = 3x^2$, $\Rightarrow du/dx = 6x$ and $v = 2x$, $\Rightarrow dv/dx = 2$

By formula, $dy/dx = du/dx + dv/dx = 6x + 2$

Question2:

$y = 3x^{1/2} + 1/\sqrt{x}$, find dy/dx

Working:

$u = 3x^{1/2}$, $\Rightarrow \frac{du}{dx} = \frac{3x^{-1/2}}{2}$, $v = 1/\sqrt{x} = 1/x^{1/2} = x^{-1/2}$, $\Rightarrow \frac{dv}{dx} = -\frac{1}{2}x^{-3/2}$

By formula, $\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx} = \frac{3x^{-1/2}}{2} + (-\frac{1}{2}x^{-3/2}) = \frac{3}{2x^{1/2}} - \frac{1}{2x^{3/2}}$

Question 3:

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$y = \frac{x^4 - 3x^3 - 4x^2 + 5}{x^2}$, find y^1.

Working:

Divide each term on the right hand side by x^2

$y = \frac{x^4}{x^2} - \frac{3x^3}{x^2} - \frac{4x^2}{x^2} + \frac{5}{x^2} = x^2 - 3x - 4x + 5x^{-2}$

Differentiating directly, $y^1 = 2x - 3 - 0 - 10x^{-3} = 2x - 3 - 10/x^3$

2. Product Of Differentiation:

$y = uv$. Where u = function of x, v = function x

thus; by formula, $\frac{dy}{dx} = \frac{udv}{dx} + \frac{vdu}{dx}$

Question 1:

$y = (2x + 1)(x - 2)$, find dy/dx

Working:

$y = (2x + 1)(x - 2)$

let $u = (2x + 1)$, ***this implies that*** , $du/dx = 2$ and $v = x - 2$,

this implies that, $dv/dx = 1$.

By formula, $\frac{dy}{dx} = \frac{udv}{dx} + \frac{vdu}{dx} = (2x+1)(1) + (x-2)(2)$

therefore, $dy/dx = 4x - 3$

Question 2:

$y = (x^2 - 2)(1 + x^{½})$, find dy/dx

Working:

Answer $= \frac{5x^{3/2}}{2} - \frac{1}{2x^{½}} + 2x$

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3. Quotient Rule (division):

If $y = \frac{u}{v}$, *then*; $\frac{dy}{dx} = \frac{v\,du/dx - u\,dv/dx}{v^2}$

Question1:

$y=(2x+1)/(x^2+3)$, find dy/dx

Working:

$u=2x+1$, ***this implies that***, $du/dx=2$, and $v=x^2+3$

this implies that, $dv/dx=2x$

By formula, $\frac{dy}{dx} = \frac{v\,dy/dx - u\,dv/dx}{v^2} = \frac{(x^2+3)(2)-(2x+1)(2x)}{(x^2+3)^2}$

$$= \frac{-2x^2-2x+6}{x^4+6x^2+9}$$

Question 2:

$y = \frac{x^2-2x}{x^3+1}$, ***then*** y^1 is --------

Working: Solve this on your own (Get Serious).

4. Function Of Functions:

By formula, $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

Question1:

$y=(2x+1)^5$, then y^1 is ----------

Working:

Method 1:

$y=(2x+1)^5$

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$u = 2x + 1$. ***this implies that*** $du/dx = 2$

thus, $y = u^5$, ***hence***, $dy/du = 5u^4$

By formula, $dy/dx = dy/du \times du/dx = 5u^4 \times 2 = 10u^4$

Recall that $u = 2x + 1$. ***therefore***, $dy/dx = 10(2x+1)^4$

Method 2:

Using short cut, $y^1 = 5(2x+1)^4 \times 2 = 10(2x+1)^4$

Question 2:

$y = \sqrt{2x+1}$, find y^1

Working:

$y = (2x+1)^{½}$. Using short cut, $dy/dx = ½(2x+1)^{-½} \times 2$

$$\therefore \frac{dy}{dx} = \frac{1}{(2x+1)^{½}}$$

ASSIGNMENT:

i. $y = (3x+7)^{1/3}$

ii. $y = \sqrt{\frac{x}{1-x}}$

iii. $y = (x^2 - 2x + 5)^5$

Derivative of Trigonometry:

i. If $y = \sin x$, then $dy/dx = \cos x$

ii. $y = \cos x$, $dy/dx = -\sin x$

iii. $y = \tan x = \sin x / \cos x$

i.e $y = \frac{\sin x \longrightarrow u}{\cos x \longrightarrow v}$

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$$\frac{dy}{dx} = \frac{v\,du/dx - u\,dv/dx}{v^2}$$

Where; $du/dx = \cos x$, $dv/dx = \sin x$

$$\Rightarrow \frac{dy}{dx} = \frac{\cos x(\cos x) - \sin x(-\sin x)}{\cos^2 x} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x}$$

From $\cos^2 x + \sin^2 x = 1$, $\Rightarrow \frac{dy}{dx} = \frac{1}{\cos^2 x} = \sec^2 x$

OR

$$\frac{\cos^2 x}{\cos^2 x} + \frac{\sin^2 x}{\cos^2 x} = 1 + \tan^2 x = \sec^2 x$$

Question 1:

$y = \sin 2x$, find y^1.

Working: $u = 2x$, $\Rightarrow du/dx = 2$, $y = \sin u$, $\Rightarrow dy/ddu = \cos u$

By formula, $dy/dx = dy/du \times du/dx = \cos u \times 2 = 2\cos u$

therefore, $dy/dx = 2\cos 2x$

Question 2:

$Y = \sin^2 x$, find dy/dx

Working: $u = \sin x$, $\Rightarrow du/dx = \cos x$, ***thus***, $y = u^2$, $\Rightarrow dy/du = 2u$

But $dy/dx = dy/du \times du/dx = 2u \times \cos x = 2u\cos x = 2\sin x \cos x$

Question 3:

$y = \sin^3 x$, find dy/dx.

Working: **Method 1:** $u = \sin x$, $\Rightarrow dy/dx = \cos x$, ***thus***, $y = u^3$,

$\Rightarrow dy/du = 3u^2$. But $dy/dx = dy/du \times du/dx = 3u^2 \times \cos x = 3u^2\cos x$

$= 3\sin^2 x \cos x$

Method 2:

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Solving directly, we have $dy/dx = 3\sin^2 x\cos x$ (thesame as above)

Question 4:

Differentiate w.r.t. x, $y = \cos(4x+5)$

Working: $u = 4x+5$, $\Rightarrow du/dx = 4$, $y = \cos u$, $\Rightarrow dy/du = -\sin u$

But $dy/dx = dy/du \times du/dx = -\sin u \times 4 = -4\sin u = -4\sin(4x+5)$

Question 5:

$y = \tan\sqrt{1-x}$, find y'

Working:

$u = \sqrt{1-x} = (1-x)^{\frac{1}{2}}$. ***thus***, $du/dx = \frac{1}{2}(1-x)^{-\frac{1}{2}} \times -1 = \dfrac{-1}{2(1-x)^{\frac{1}{2}}}$

i.e $y = \tan u$, $\Rightarrow dy/du = \sec^2 u$

but $dy/dx = dy/du \times du/dx = \sec^2 u \times \dfrac{-1}{2(1-x)^{\frac{1}{2}}} = -\sec^2 u$

$$\therefore \frac{dy}{dx} = \frac{-\sec^2\sqrt{1-x}}{2\sqrt{1-x}}$$

Question 6:

$y = x\sin^4 x$, find dy/dx

Working:

$y = x\sin^4 x$

$u = x$, $\Rightarrow du/dx = 1$ and $v = \sin^4 x$, $dv/dx = 4\sin^3 x\cos x$

But $dy/dx = u\,dv/dx + v\,du/dx = (x)(4\sin^3 x\cos x) + (\sin^4 x)(1)$

$= 4x\sin^3 x\cos x + \sin^4 x$

NB:

y	y'
(1). sinax	acosax

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2.	cosax	-asinax
3.	$\sin x^2$	$2x\cos x^2$
4.	sinx/4	¼ cosx/4
5.	$x\sin^4 x$	$4x\sin^3 x\cos x + \sin^4 x$

Differentiation Of Exponential Functions:

If $y = e^x$, *then*, $dy/dx = e^x$

OR

$y = e^x$. Let $u = x$, => du/dx 1, $y = e^u$, => $dy/du = e^u$

But $dy/dx = dy/du \times du/dx = e^u \times 1 = e^u = e^x$

Question 1:

$y = e^{3x^2}$, find y^1

Working: Differentiating directly, $dy/dx = 6xe^{3x^2}$

Question 2:

$y = e^t \sin t$

Working:

$y = e^t \sin t$, $u = e^t$, $du/dt = e^t$, $v = \sin t$, $dv/dt = \cos t$

But $dy/dt = u\,dv/dt + v\,du/dt = (e^t)(\cos t) + (\sin t)(e^t) = e^t(\cos t + \sin t)$

Question 3:

$y = (e^x - e^{-x})^2$, find dy/dx

Working: Solving using short cut,

$dy/dx = 2(e^x - e^{-x}) \times (e^x - (-e)^{-x}) = (2e^x - 2e^{-x})(e^x + e^{-x})$

$= 2e^{2x} + 2e^{x-x} - 2e^{-x+x} - 2e^{-x-x} = 2e^{2x} + 2 - 2 - 2e^{-2x}$

$\therefore dy/dx = 2(e^{2x} - e^{-2x})$

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NB:

y	y^1
1. e^x	e^x
2. e^{ax}	ae^{ax}
3. e^{3x}	$3e^{3x}$
4. e^{-x}	$-e^{-x}$
5. e^{x2}	$2xe^{x^2}$

Differentiation Of Logarithms Functions:

1. $y = \log_e x$ or $\ln x$, ***then*** $dy/dx = 1/x$
2. $y = \log_a x$, ***then*** $dy/dx = \log_a x \times \log_a e = \frac{1}{x} \log_a e$

Question 1:

$y = \log_e x$, find dy/dx

Working:

$y = \log_e x$; $e^y = x$; i.e $e^y dy/dx = 1$

$$\frac{dy}{dx} = \frac{1}{e^y} = \frac{1}{x}$$

Question 2:

$y = \log_e \sin x$, find y^1.

Working:

$y = \log_e \sin x$; $e^y = \sin x$; i.e $e^y \frac{dy}{dx} = \cos x$

$$\frac{dy}{dx} = \frac{\cos x}{e^y} = \frac{\cos x}{\sin x}$$. ***therefore***, $dy/dx = \cot x$

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Question 3:

$y = \ln \sin x/3$, find dy/dx

Working: Solving using short cut, $\frac{dy}{dx} = \frac{1/3 \cos x/3}{\sin x/3} = \frac{1}{3} \cot \frac{x}{3}$

Question 4:

$y = \log_e x$, find y'.

Working: Solving directly, $y' = 1/x \log_e e$

Question 5:

$y = \log_e x^2$

Working: Solving directly, $\frac{dy}{dx} = \frac{1}{x^2} \log_e e \times 2x = \frac{2}{x} \log_e e$

Differentiation Of General Exponential Functions:

$y = a^x$

Take In of both sides

$\ln y = \ln a^x$

$\frac{1}{y} \frac{dy}{dx} = \ln a$; $\frac{dy}{dx} = y \ln a = a^x \ln a$

Question 1:

$y = 3^x$, find dy/dx.

Take In of both sides

$\ln y = \ln 3^x$

$\ln y = x \ln 3$

$\frac{1}{y} \frac{dy}{dx} = \ln 3$ i.e $\frac{dy}{dx} = y \ln 3 = 3^x \ln 3$

Question 2:

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$y = \log_e \frac{x}{x-1}$, find dy/dx.

Working: $\log_e \frac{x}{x-1} = \ln \frac{x}{(x-1)}$

$\Rightarrow y = \ln x - \ln(x-1)$

$\frac{dy}{dx} = \frac{1}{x} - \frac{1}{x-1} = \frac{-1}{x-1}$

Question 3:

$y = \log_e \frac{x^2}{\sqrt{x^2-1}}$, find dy/dx.

Working:

$y = \log_e \frac{x^2}{\sqrt{x^2-1}} = \ln \frac{x^2}{(x^2-1)^{1/2}} = \ln x^2 - \ln(x^2-1)^{1/2}$

$\frac{dy}{dx} = \frac{2}{x} - \frac{\frac{1}{2}(x^2-1)^{-1/2} \times 2x}{(x^2-1)^{1/2}}$

$= \frac{2}{x} - \frac{x}{(x^2-1)^{1/2} \times (x^2-1)^{1/2}} = \frac{2}{x} - \frac{x}{x^2-1} = \frac{x^2}{x(x^2-1)}$

Higher Derivatives:

Question 1:

Given, $y = 4x^3$, $dy/dx = 12x^2$, $d^2y/dx^2 = 24x$, $d^3y/dx^3 = 24$

Question 2: If $y = 2x^5 - x^3 + 2x - 4$, ***then***, $y^1 = 10x^4 - 3x^2 + 2$

$y^{11} = 40x^3 - 6x$, $y^{111} = 120x^2 - 6$, $y^{iv} = 240x$, $y^v = 240$

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Implicit Functions:

Question 1:

Find dy/dx from the expression, $2x^2y^3 - 6xy^2 + 4x^5$

Working:

This is a combination of implicit function and product rule.

Note, we treat this as a product rule with x and y variable, but anytime we differentiate function of y we multiply by $\frac{dy}{dx}$

Now let's go!

For $2x^2y^3$, apply udv + vdu, where $u = 2x^2$ and $v = y^3$. ***thus***,

$$2x^2y^3 = 2x^2(3y^2)\frac{dy}{dx} + y^3(4x)$$

For $6xy^2$, apply udv + vdu, ***thus***, $6xy^2 = 6x(2y)\frac{dy}{dx} + y^2(6)$

For $4x^5 = 20x^4$. Combining the 3, we have

$$2x^2(3y^2)\frac{dy}{dx} + y^3(4x) - \left(6x(2y)\frac{dy}{dx} + y^2(6)\right) + 20x^4$$

$$6x^2y^2\frac{dy}{dx} + 4xy^3 - 12xy\frac{dy}{dx} - 6y^2 + 20xy$$

$$6x^2y^2\frac{dy}{dx} - 12xy\,dy + 4xy^3 - 6y^2 + 20x^4$$

$$\frac{dy}{dx}(6x^2y^2 - 12xy) + 4xy^3 - 6y^2 + 20x^4$$

Question 2:

Find dy/dx from the equation $x^2 - y^2 + 3x = 2y$

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Working:

$x^2 - y^2 + 3x = 2y$

$2x - 2y\frac{dy}{dx} + 3 = 2\frac{dy}{dx}$

Collect like terms of dy/dx

$2x + 3 = 2dy/dx + 2ydy/dx$

$2x + 3 = \frac{dy}{dx}(2 + 2y)$; *thus*, $\frac{dy}{dx} = \frac{2x+3}{2+2y}$

Question 3:

Differentiate w.r.t. x; $x^2 + 2xy + 3y^2 = 4$, at $x = 3$ and $y = 2$

Working:

$x^2 + 2xy + 3y^2 = 4$

$2x + 2y + 2x\,dy/dx + 6ydy/dx = 0$

$2xdy/dx + 6ydy/dx = -2x - 2y$; $dy/dx\,(2x+6y) = -2x - 2y$

$\frac{dy}{dx} = \frac{-2x-2y}{2x+6y}$

At $x = 3, y = 2$, we have ; $\frac{dy}{dx} = \frac{-2(3)-2(2)}{2(3)+6(2)} = \frac{-5}{9}$

Question 4:

Working:

$4y^2x - 5x^2y^3 + 4y = 0$

$8yxdy/dx + 4y^2 - 10xy^3 - 15x^2y^2dy/dx + 4dy/dx = 0$

Collect like terms of dy/dx

$8yxdy/dx - 15x^2y^2dy/dx + 4dy/dx = 10xy^3 - 4y^2$

$dy/dx\,(8yx - 15x^2y^2 + 4) = 10xy^3 - 4y^2$

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$$\frac{dy}{dx} = \frac{10xy^3 - 4y^2}{8yx - 15x^2y^2 + 4}$$

Exercise:

Find the dy/dx from the following equations:

(1). $(x+y)^2 = 5$ (2). $x^2y + y^2x + 4x = 1$ (3). $x^2y^3 = 25y$

APPLICATION OF DIFFERENTIATION:

I. Tangent To A Curve:

Question 1:

Find the tangent to the curve; $y = 2x^3 - x^2 + 3x + 1$ at $x = 1$

Working:

By definition, the tangent to a curve or circle is the $\frac{dy}{dx}$ of the curve or circle, and the equation of such tangent at any point (x, y) is got from

$y - y_1 = \frac{dy}{dx}(x - x_1)$ or $y - y_1 = m(x - x_1)$

Given $y = 2x^3 - x^2 + 3x + 1$

Let's find dy/dx. i.e $dy/dx = 6x^2 - 2x + 3$. At $x = 1$,

$dy/dx = 6(1)^2 - 2(1) + 3 = 7$

Question 2:

Find the equation of the tangent to the curve; $x^2y + y^3x + 3x - 13 = 0$, at the point (1,2)

Working:

$2xy + x^2dy/dx + 3y^2xdy/dx + y^3 + 3 - 0 = 0$

$x^2dy/dx + 3y^2xdy/dx = -y^3 - 2xy - 3$

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$dy/dx\,(x^2+3y^2x) = -y^3 - 2xy - 3$

$$\frac{dy}{dx} = \frac{-y^3 - 2xy - 3}{x^2+3y^2x}$$

at the point (1,2)

$$\frac{dy}{dx} = \frac{-2^3 - 2(1)(2) - 3}{1^2+3(2)^2(1)} = \frac{-15}{13}$$

Where slope = Gradient = $m = dy/dx = -15/13$

But equation of tangent is given by;

$y - y_1 = m(x - x_1)$; $y - 2 = -15/13\,(x-1)$; $13(y-2) = -15(x-1)$

therefore, $15x + 13y - 41 = 0$

2. *Normal To A Curve:*

QUESTION 1:

Find the equation of the normal to the curve

$9x^2 - 4y^2 = 108$, at the point (4,3)

Working:

By definition, the gradient of the normal$(m_2) = -1/m_1$

Where m_1 = gradient of the tangent. given, $9x^2 - 4y^2 = 108$

i.e $18x - 8ydy/dx = 0$

thus, $\frac{dy}{dx} = \frac{18x}{8y}$

at the point (4,3), $\frac{dy}{dx} = \frac{18(4)}{8(3)} = \frac{72}{24} = 3$

Hence, $dy/dx = m_1 = 3$

But $m_1 m_2 = -1$

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$= m_2 = -1/m_1 = -1/3$

By formula, the equation of a normal is given by: $y - y_1 = m_2(x - x_1)$

i.e $y - 3 = -1/3\,(x - 4)$

therefore, $3y + x - 13 = 0$

3. Velocity And Acceleration:

Velocity (v) measures the rate of change of distance (s) with respect to time(t). ***thus,*** Velocity (v) $= \frac{ds}{dt}$

SIMILARLY, acceleration a, measures the rate of change of velocity with respect to time. ***thus,*** Acceleration, $a = \frac{dv}{dt} = \frac{d^2s}{dt^2}$

Question 1:

The distance traveled by a particle in a straight line in t seconds is given by the formula $s = 2t + t^2$, calculate (1). its velocity after 3 seconds (2). The acceleration of the particle

Working:

(1). Given $s = 2t + t^2$

thus, $v = ds/dt = 2 + 2t$. For $t = 3$, velocity, $v = 2 + 2(3) = 8$units/sec

(2). $a = dv/dt = 2$ units/sec^2

Question 2:

The distance s, of a moving particle from a certain fixed point P is given by the formula, $s = t^3 - 12t^2 + 36t + 2$ (meters)

i. Calculate the distance of the particle from P when $t = 0$?

ii. Calculate its velocity at $t = 0$?

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iii. When is the particle stationary

iv. What is its acceleration after 4 seconds

Working:

i. $s=t^3-12t^2+36t+2$, when $t=0$, we have $s=2m$

ii. $v=ds/dt=3t^2-24t+36$, at $t=0$, $v=3(0)^2-24(0)+36$

thus, $v=36$ m/sec

iii. The particle is stationary when $ds/dt=v=0$

i.e. $3t^2-24t+36=0$

Factorizing, we have; $t=2$secs or 6secs

iv. $a=dv/dt=6t-24$, at $t=4$, $\Rightarrow$ acc(a) $=6(4)-24=0$

$\therefore a=0m/sec^2$

4. *Maximum And Minimum Point:*

Note the following;

(I) If $d^2y/dx^2<0$, we have Maximum point

(ii). If $d^2y/dx^2=0$, we have point of inflexion

(iii). If $d^2y/dx^2>0$, we have minimum point.

Question1:

Let $y=x^2-4x+3$, find the turning point and determine whether it is a maximum or minimum point.

Working:

$y=x^2-4x+3$. ***thus***, $dy/dx=2x-4$

At the turning point, $dy/dx=0$.

hence, $dy/dx=2x-4=0$ i.e. $x=2$.

thus, at the turning point, $x=2$

But $d^2y/dx^2=2$, at $x=2$, $d^2y/dx^2=2>0$(we have minimum point

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Question 2:

Given the function; f: x $\longrightarrow x^3-6x^2+9x+2$ (x being real)

(i) find the stationary point

(ii) find the nature of these stationary points

Working:

(i). $f(x)=y=x^3-6x^2+9x+2$

thus, $dy/dx=3x^2-12x+9$

At the stationary point, $dy/dx=0$ i.e. $3x^2-12x+9=0$, i.e $x=1$ or 3.

Hence, at the stationary point, $x=1$ or 3

(ii). Recall that; $dy/dx=3x^2-12x+9$

So, $d^2y/dx^2=6x-12$. At $x=1$, we have $d^2y/dx^2=6(1)-12=-6$ (max.pt)

Again, at $x=3$, $d^2y/dx^2=6(3)-12=6$ (min. pt.)

thus, at $x=1$, (max point) and at $x=3$ (minimum point)

Question 3:

Find the point of inflexion on the curve, $y=x^3-6x^2+11x-4$.

Working:

Given; $y=x^3-6x^2+11x-4$

NB: for point of inflexion, $d^2y/dx^2=0$

So, $dy/dx=3x^2-12x+11$

$d^2y/dx^2=6x-12=6x-12=0$, ***thus***, $x=2$

∴ At the point of inflexion, $x=2$

Concept Of Integration:

Generally, if $y=x^n$, then $\int dy = \frac{x^{n+1}}{n+1} + C$

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Where C = Constant

Question 1:

Evaluate the following

(I) $\int x^4 dx$ (ii). $\int 3x^3 dx$

Working:

i. $\int x^4 dx = \frac{x^5}{5} + C$ (ii). $\int 3x^3 dx = \frac{3x^4}{4} + C$

Question 2:

i. Integrate $y = (5x+1)^6$

Working:

$\int dy = \int (5x+1)^6 dx$; Let $u = 5x + 1$, $\Rightarrow dx = du/5$

thus, $\int dy = \int u^6 . du/5 = 1/5 \int u^6 . du$

$$y = \frac{1}{5}\left(\frac{u^7}{7}\right) + C = \frac{u^7}{35} + C = \frac{(5x+1)^7}{35} + C$$

ii. Evaluate $\int x(2x^2+1)^7 dx$

Working:

$$\int dy = \int x(2x^2+1)^7 dx$$

Let $u = 2x^2 + 1$, this implies that $du/dx = 4x$; $dx = du/4x$

i.e $\int dy = \int x.u^7 \frac{du}{4x} = \frac{1}{4}\int u^7 du$; $y = \frac{1}{4}\left(\frac{u^8}{8}\right) + C = \frac{(2x^2+1)^8}{32} + C$

(iii). Evaluate $\int x^2(5x^3+1)^4 dx$

Working:

$$\int dy = \int x^2 (5x^3+1)^4 dx$$

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Working:

Let $u = 5x^3 + 1$, $\Rightarrow du/dx = 15x^2$ i.e $dx = du/15x^2$

i.e $\int dy = \int x^2 . u^4 \frac{du}{15x^2} = \frac{1}{15}\int u^4 . du$; $y = \frac{1}{15}\left(\frac{u^5}{5}\right) + C$

$= \frac{(15x^3+1)^5}{75} + C$

Integration Of Exponetial Functions:

By formula, if $y = e^x dx$, ***then***, $\int y = e^x + C$

Question1:

Evaluate. $\int e^{2x} dx$

Working:

$\int dy = \int e^{2x} dx$. Let $u = 2x$, ***thus***, $du/dx = 2$ i.e $dx = du/2$

thus, $\int dy = \int e^u . du/2 = \frac{1}{2}\int e^u du = \frac{1}{2} xe^u + C$. ***therefore***, $y = \frac{e^{2x}}{2} + C$

Question 2:

Evaluate $\int xe^{x^2} dx$

Working:

$\int dy = \int xe^{x^2} dx$, let $u = x^2$. ***thus***, $du/dx = 2x$. i.e $dx = du/2x$

$\int dy = \int xe^u \frac{du}{2}$; $y = \frac{1}{2}\int e^u du = \frac{1}{2} e^u + C = \frac{e^{x^2}}{2} + C$

Integration Of The Form:

$$\int \frac{f'(x)}{f(x)} = \ln f(x) + 1$$

Question 1:

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Evaluate $\int 1/x$

Working: $\int 1/x = \ln x + C$

Question 2:

Evaluate $\int \frac{1}{x+1}\,dx$

Working:

Solving directly, ***answer*** $= \ln(x+1) + C$

Question 3: Evaluate $\int 5/(x+1)\,dx$

Working: . Let $u = x+1$, $\Rightarrow dx = du$

thus, $\int 5/(x+1)dx = \int 5/u\,du = 5\int 1/u\,du = 5\ln u + C = 5\ln(x+1) + C$

Question 4:

Evaluate $\int \frac{3x}{2x^2+1}\,dx$

Working:

$$\int \frac{3x}{2X^2+1}\,dx = \frac{3}{4}\ln(2x^2+1) + C$$

Question 5:

EVALUATE $\int \frac{5x^2}{4x^3+1}\,dx$

Working: $\int \frac{5x^2}{4x^3+1} = \frac{5}{12}\ln(4x^3+1) + C$

Question 6:

Evaluate $\int \frac{1}{x^2-5x+6}\,dx$

Working:

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$$\int \frac{1}{x^2-5x+6}\,dx = \int \frac{1}{(x-2)(x-3)}\,dx \equiv \int \frac{A}{x-2} + \frac{B}{x-3}\,dx$$

$1 \equiv A(x-3)+B(x-2)$, if $x=3$, we have $1=A(0)+B(3-2)$, $\Rightarrow B=1$

Also, if $x=2$, i.e. $1=A(2-3)+0$. ***therefore***, $A=-1$

thus,

$$\int dy = \int \frac{-1}{x-2} + \frac{1}{x-3}\,dx$$

On integration, $y=-\ln(x-2)+\ln(x-3)+C$

Question 7:

Evaluate $\int \frac{2x+1}{x^2+5x+4}\,dx$

Ans is: $\frac{7}{3}\ln(x+4) - \frac{1}{3}\ln(x+1)+C$

Question 8:

Evaluate $\int \frac{x^2+x+1}{x+1}\,dx$

Working:

$$\begin{array}{r|l} & x \\ \hline x+1 & x^2+x+1 \\ & x^2+x \\ \hline & -\ 0+1 \end{array}$$

i.e. $\int \frac{x^2+x+1}{x+1}\,dx = \int x + \frac{1}{x+1}\,dx$

On integration, we have; $\frac{x^2}{2}+\ln(x+1)+C$

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Question 9:

Evaluate $\int \frac{3x^2+2x+1}{2x+4}\,dx$

Working:

$$\begin{array}{r|l} & \frac{3}{2}x - 2 \\ \hline 2x+4 & 3x^2+2x+1 \\ & -\ 3x^2+6x \\ \hline & -\begin{pmatrix} -4x+1 \\ -4x-8 \end{pmatrix} \\ \hline & 9 \end{array}$$

$$\Rightarrow \int \frac{3x^2+2x+1}{2x+4}\,dx = \int \frac{3}{2}x - 2 + \frac{9}{2x+4}\,dx$$

On integration, we have $\frac{3x^2}{4} - 2x + \frac{9}{2}\ln(2x+4) + C$

Integration Of Trigonometric Functions :

NB:

1. $\int \sin x\,dx = -\cos x + C$ (2). $\int \cos x\,dx = \sin x + C$
3. $\int \tan x\,dx = -\ln\cos x + C$ (4). $\int \sec^2 x\,dx = \tan x + C$

Question 1:

Evaluate $\int \cos 5x\,dx$

Working:

Let $u = 5x, \Rightarrow dx = du/5$. *thus*, $\int \cos 5x\,dx = \int \frac{\cos u\,du}{5}$

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$=1/5\int \cos u\,du$, on integration, we have $1/5 \sin u + C = 1/5 \sin 5x + C$

Question 2:

$\int \sin 6x = -1/6 \cos 6x + C$ (i.e use thesame procedure as above)

Question 3:

Evaluate $\int \sin^2 x\,dx$

Working:

NB: (1). $\sin^2 x = 1 - \cos^2 x$ (2). $\sin^2 x = \frac{1-\cos 2x}{2}$

3. $\cos^2 x = 1 - \sin^2 x$ (4). $\cos^2 x = \frac{1+\cos 2x}{2}$

5. $2\sin A\cos B = \sin(A+B) + \sin(A-B)$
6. $2\cos A\sin B = \sin(A+B) - \sin(A-B)$
7. $2\cos A\cos B = \cos(A+B) + \cos(A-B)$
8. $2\sin A\sin B = \cos(A+B) - \cos(A-B)$

So, $\int \sin^2 x\,dx = \int \frac{1-\cos 2x}{2}\,dx = \int ½\,dx - ½\int \cos 2x\,dx$

On integration, we have $½x - ¼\sin 2x + C$

Question 4:

Evaluate $\cos^2 x\,dx$

Working:

$\int \cos^2 x\,dx = \int \left(\frac{1+\cos 2x}{2}\right) dx = \int ½\,dx + ½\int \cos 2x\,dx$

$= ½x + ¼\sin 2x + C$

Question 5:

Evaluate $\int \sin^3 x\,dx$

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Working:

$$\int \sin^3 x\,dx = \int \sin^2 x.\sin x\,dx = \int (1-\cos^2 x).\sin x\,dx$$

$$= \int \sin x\,dx \int \cos^2 x \sin x\,dx$$

Let $u = \cos x$, ***this implies that*** $\frac{du}{dx} = -\sin x$, *thus*, $dx = \frac{-du}{\sin x}$

Now, we have;

$$\int \sin x \cos^2 x\,dx = \int \sin x.u^2 - \frac{du}{\sin x} = -\int u^2 du = \frac{-u^3}{3} = \frac{-\cos^3 x}{3}$$

$\int \sin x\,dx = -\cos x$

thus, $\int \sin x\,dx - \int \sin x \cos^2 x\,dx = -\cos x + \frac{\cos^2 x}{3} + C$

Question 6:

Evaluate $\int \cos^3 x\,dx$

Working:

$$\int \cos^3 x\,dx = \sin x - \frac{\sin^3 x}{3} + C$$

Question 7:

Evaluate $\int \sin^4 x\,dx$

Working:

$$\int \sin^4 x\,dx = \int (\sin^2 x)^2 dx = \int \left(\frac{1-\cos^2 x}{4}\right)^2 dx$$

NB: $\cos^2 x = \frac{1}{2}(1+\cos 2x)$, $\Rightarrow \cos^2 2x = \frac{1}{2}(1+\cos 4x)$

thus, $\int \left(\frac{1-\cos 2x}{4}\right)^2 dx = \int \left(\frac{1-2\cos 2x+\cos^2 2x}{4}\right) dx$

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$= \frac{1}{4}\int(1-2\cos 2x+\frac{1}{2}+\frac{1}{2}.\cos 4x)dx = \frac{1}{4}\int(3/2\,x-2\sin 2x+\frac{1}{2}.\cos 4x)\,dx$

$$= \frac{1}{4}\left[\frac{3}{2}x-2\sin 2x+\frac{1}{2}\frac{\sin 4x}{4}\right]+C = \frac{3}{8}x-\frac{\sin 2x}{4}+\frac{\sin 4x}{32}+C$$

Question 8: Evaluate $\int \cos^4 x\,dx$

Working:

$$\int \cos^4 x dx = \frac{3}{8}+\frac{\sin 2x}{4}-\frac{\sin 4x}{32}+C$$

Question 9: Evaluate $\int \sin 4x.\cos 2x dx$

Working:

Using $2\sin A\cos B = \sin(A+B)+\sin(A-B)$, *thus,*

$\sin 4x.\cos 2x = \frac{1}{2}(2\sin 4x\cos 2x) = \frac{1}{2}\{\sin(4x+2x)+\sin(4x-2x)\}$

$= \frac{1}{2}(\sin 6x+\sin 2x)$, ***thus,*** $\int \sin 4x\cos 2x dx = \frac{1}{2}\int(\sin 6x+\sin 2x)dx$

On integration, we have $\frac{-\cos 6x}{12}-\frac{\cos 2x}{4}+C$

Question 10:

Evaluate $\int \cos 5x\sin 3x dx$

Working:

$\int \cos 5x\sin 3x dx = \frac{1}{2}\int(2\cos 5x\sin 3x)dx$

i.e $\frac{1}{2}\int\{\sin(5x+3x)-\sin(5x-3x)\}dx = \frac{1}{2}\int(\sin 8x-\sin 2x)dx$

$$= \frac{1}{2}\left(\frac{-\cos 8x}{8}+\frac{\cos 2x}{2}+C\right) = \frac{\cos 2x}{4}-\frac{\cos 8x}{16}+C$$

Question 11: Evaluate $\int \cos 6x\cos 4x dx$

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Working:

Answer $= \frac{\sin 10x}{20} + \frac{\sin 2x}{4} + C$

Question 12: $\int \sin 5x \sin x\,dx$

Answer $= \frac{\sin 4x}{8} - \frac{\sin 6x}{12} + C$

More On Integration Of Exponential Functions:

Note that $\int e^x dx = e^x + C$

Question 1:

Find $\int e^{5x} dx$

Working:

Let $u = 5x$, $\Rightarrow dx = du/5$. ***thus,*** $\int e^{5x} dx = \int e^u . du/5 = 1/5 \int e^u . du$

$= \frac{1}{5} e^u + C = \frac{e^{5x}}{5} + C$

Question 2: Evaluate $\int e^{5x} + 4\,dx$

Working: ***Answer*** $= \frac{e^{5x}}{5} + 4 + C$

Question 3: $\int e^x (e^x + 1) dx$

Working:

Let $u = e^x + 1$. ***this implies that*** $du/dx = e^x$, ***thus,*** $= dx = du/e^x$

i.e. $\int e^x (e^x + 1) dx = \int e^x . u . du/e^x = \int u du = \frac{u^2}{2} + C = \frac{(e^x + 1)^2}{2} + C$

Question 4:

Evaluate $\int x e^{x^2} dx$

Working: ***Answer*** $= \frac{1}{2} e^{x^2} + C$

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Integration by parts:

Generally, ***by formula***, $\int udv = uv - \int vdu$

Question 1:

Evaluate $\int x \ln x dx$

Working:

Let $u = \ln x$, **and** $dv = xdx$

i.e. $du/dx = 1/x$, $\Rightarrow du = 1/x dx$ **and** $\int dv = \int xdx$, ***thus***, $v = x^2/2$

By formula, $\int udv = uv - \int vdu$

Substituting, we have;

$$\int udv = \ln x \frac{x^2}{2} - \int \frac{x^2}{2} \cdot \frac{1}{x} dx = \frac{x^2 \ln x}{2} - \frac{1}{2}\int x dx$$

$$= \frac{x^2 \ln x}{2} - \frac{1}{2} \cdot \frac{x^2}{2} + C = \frac{x^2}{2}\left(\ln x - \frac{1}{2}\right) + C$$

Question 2:

Evaluate $\int x^2 \ln x dx$. ***Working: Answer*** $= \frac{x^3}{3}\left(\ln x - \frac{1}{3}\right) + C$

Question 3: $\int xe^{2x} dx$

Working:

Let $u = x$, i.e $du = dx$ **and** $dv = e^{2x}$, ***thus***, $v = \frac{e^{2x}}{2}$

But $\int udv = uv - \int vdu = (x)\frac{e^{2x}}{2} - \int \frac{e^{2x}}{2} \cdot dx = \frac{xe^{2x}}{2} - \frac{1}{4}e^{2x} + C$

$$= \frac{e^{2x}}{2}\left(\frac{x-1}{2}\right) + C$$

Question 4: Evaluate $\int x \cos x dx$

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Working:

Let $u = x$, i.e $du = dx$ and $dv = \cos x dx$. ***thus*** , $\int dv = \int \cos x dx = \sin x$

But $\int udv = uv - \int vdu = x\sin x - \int \sin x dx = x\sin x + \cos x + C$

Question 5:

Evaluate $\int x\sin 2x dx$

Working: Answer $= \frac{-\cos 2x}{2} + \frac{\sin 2x}{4} + C$

NB:

1. If one factor is a log function, that must be taken as 'u'
2. If there is no log function, but a power of x, that becomes 'u'
3. If there is neither a log function nor a power of x, then the exponential function is taken as 'u'

So, which would you choose as u in the following cases;

1. $\int x^4 \cos 2x dx$, u= ----------
2. $\int x^4 e^{3x} dx$, u=----------
3. $\int x^3 \ln(x+4) dx$, u=----------
4. $\int e^{2x} \cos 4x dx$, u=----------

ANSWERS:

1. $u = x^4$ (2). $u = x^4$
3. $u = \ln(x+4)$ (4). $u = e^{2x}$

DEFINITE INTEGRAL:

Question 1: Evaluate $\int_0^1 (2x-1)^5 dx$

Working:

$\int_0^1 (2x-1)^5 dx$; let $u = 2x-1$; $\Rightarrow dx = du/2$

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So, $\int_0^1 (2x-1)^5 dx = \int_0^1 u.\frac{du}{2} = \frac{1}{2}\int_0^1 u^5 du = \frac{1}{2}\left[\frac{u^6}{6}\right]_0^1 = \frac{1}{2}\left[(2x-1)^6\right]_0^1$

Upper limit – lower limit i.e $\frac{1}{2}\{[(2(1)-1)]^6 - [(2(0)-1)]^6\} = 0$

APPLICATION OF INTEGRATION:

1. Area Under A Curve:

By formula, Area (A) $= \int_{x_1}^{x_2} y dx$

Question 1:

Find the area bounded by curve, $y = x^2 + 2$, the x – axis, the lines $x = 1$ and $x = 4$

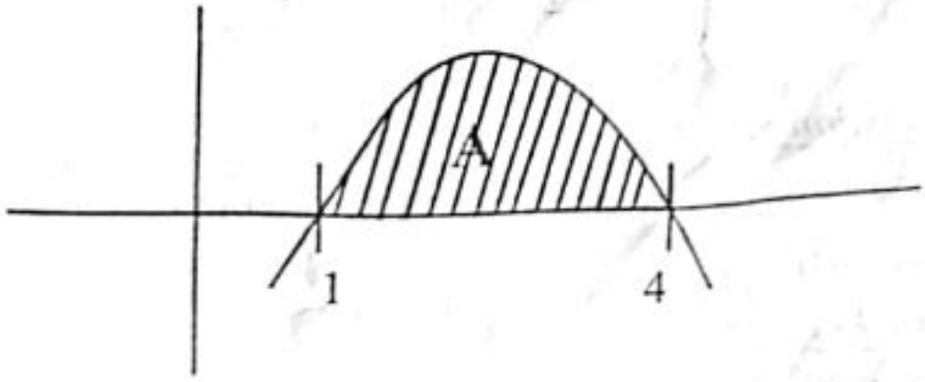
Working:

$$\text{Area, } A = \int_{x_1}^{x_2} y dx$$

Where $y = x^2 + 2, x_1 = 1, x_2 = 4$

thus, $\int_1^4 (x^2+2)dx = \left[\frac{x^3}{3} + 2x\right]_1^4 = 27$ sq. units

i.e



NB; Area being positive (i.e. 27 sq. units) here, indicate that the area

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in question is above the $x-$axis.

Question 2:

Find the area bounded by the curve, $y = x^2 - 5x+6$, and the x-axis

Working:

Whenever, you hear the x-axis, note that $y = 0$, OR you hear the y-axis, note that $x = 0$.

Now between the curve and the x-axis, it means $y = 0$,

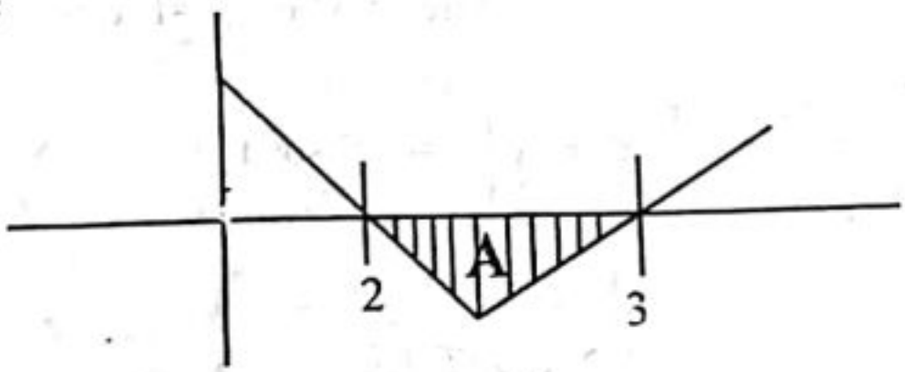
$\Rightarrow 0 = x^2 - 5x + 6$ i.e. $(x-2)(x-3) = 0$; $x = 2$ or 3

Remember, Area, $A = \int_{x_1}^{x_2} y\,dx = \int_{x_1}^{x_2} (x^2 - 5x+6)\,dx$

$= \left[\frac{x^3}{3} - \frac{5x^2}{2} + 6x\right]_2^3$; Upper limit $-$ lower limit $= -1/6$

thus, $A = 1/6$ sq. units.

i.e



Question 3:

Find the area enclosed between the curve $y = 4x(x-1)(x-2)$ and the x-axis.

Working:

At the x-axis, $y = 0$, i.e $0 = 4x(x-1)(x-2)$

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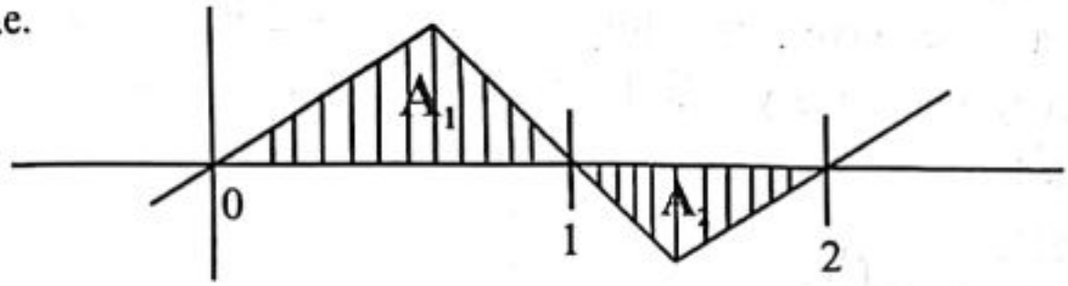
$\Rightarrow x=0,\ x=1$ and $x=2$. But, $A=\int_{x_1}^{x_2} y\,dx + \int_{x_2}^{x_3} y\,dx = A_1+A_2$

Where; $A_1=\int_0^1(4x^3-12x^2+8x)dx=1$ sq.units

$A_2=\int_{x2}^{x3} y\,dx=\int_1^2(4x^3-12x^2+8x)dx=-1=1$ sq.unit.

thus, $A_1+A_2=1+1=2$ sq.units

i.e.



2. *Volume Of Revolution:*

NB:

1. Generally, if the curve is rotated around the x-axis *then* volume,
$$v=\pi\int_{x_2}^{x_3} y^2dx$$

2. Again, if the curve is rotated around the y-axis, volume,
$$v=\pi\int_{x_2}^{x_3} x^2dy$$

Question 1:

The portion of the curve $y=2x^2$, between $x=0$ and $x=2$ is rotated completely around the x-axis. Find the volume of the solid generated.

Working:

By formula, $v=\pi\int_{x_2}^{x_3} y^2dx$, where; $y=2x^2$; i.e $y^2=(2x^2)^2=4x^4$

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$$x_1=0,\ x_2=2;\ \Rightarrow v=\pi\int_1^2 4x^4dx=4\pi\int_1^2 x^4dx=4\pi\left[\frac{x^5}{5}\right]_0^1$$

= Upper limit – Lower limit = $4\pi(32/5)=128\pi/5$ Cubic units

Question 2:

The part o the curve $y = x^2$ from $x = 1$ to $x = 2$ is rotated completely round the y-axis. Find the volume of the solid generated.

Working:

By formula, $v=\pi\int_{x_2}^{x_1}x^2dy$

Where $y=x^2$ i.e $x=y^{1/2}$, $\Rightarrow x^2=y$

Recall that $y=x^2$ .For $x=2, y=2^2=4$

thus, $v=\pi\int_{x_2}^{x_1}ydy=\pi\left[\frac{y^2}{2}\right]_1^4=\frac{15\pi}{2}$ cubic units

3. Velocity and Acceleration:

Question 1:

The velocity of a particle moving in a straight line t seconds from the starting point is given by $v=3t^2-2t+1$

(i) Find the distance in metre it covers after 30 seconds from the starting point

(ii). What is its acceleration at that time.

Working:

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Notice that velocity = $\frac{\text{Change in distance}}{\text{Change in time}}$

i.e $v = \frac{ds}{dt} = 3t^2 - 2t + 1$

$\frac{ds}{1} = (3t^2 - 2t + 1)dt$

To find distance, we integrate both sides

$\int ds = \int (3t^2 - 2t + 1)dt$

$s = \frac{3t^2}{3} - \frac{2t^2}{2} + t + C = t^3 - t^2 + t + C$; at $t=0$, $s=0$

i.e $0 = 0^3 - 0 + 0 + C$, $\therefore C = 0$

When $t=3$ seconds, $s = 33 - 32 + 3 = 21$ metres

(ii). Acceleration $= \frac{dv}{dt}$, Since $v = 3t^2 - 2t + 1$, $\frac{dv}{dt} = 6t - 2$

At time $= t = 3$, acceleration $= dv/dt = 6(3) - 2 = 16 m/s^2$

Question 2:

The velocity of a particle V which moves in a straight line is given by $V = 4 - 4\sin 2t$, if S is the distance in metre and t time in seconds, find the distance of the particle from the starting point after π seconds.

Working:

$V = \frac{ds}{dt} = 4 - 4\sin 2t$

$\int ds = \int_0^{\pi} (4 - 4\sin 2t)dt$

$S = 4t - 4 \times \frac{1}{2} \times -\cos 2t + C = [4t + \cos 2t]_0^{\pi}$

At time $t = 0 \rightarrow t = \pi$

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$S = [4\pi + 2\cos 2\pi] - [4(0) + 2\cos 0] = [4\pi + 2] - 2 = 4\pi$

(ii). Given that $dv/dt = \pi(20 - t^2)$, find V if $V = 0$ when $t = 0$

Working:

$$\int dv = \int \pi(20 - t^2)dt$$

$$V = \pi\int(20 - t2)dt = \pi\left[20t - \frac{t^3}{3}\right] + C$$

When $V = 0$, $t = 0$; i.e $0 = \pi\left[20(0) - \frac{0}{3}\right] + C$; $C = 0$

Thus, $V = \frac{\pi}{3}[60t - t^3]$

Coordinate Geometry:

Terminologies:

1. Distance(d) or Length Between Two Points:

By formula, distance $D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Question 1:

Find the distance between the point A(1,2) and B(4,6)

Working:

By formula, distance, $D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Where $x_1 = 1, x_2 = 4, y_1 = 2, y_2 = 6$

$$\Rightarrow D = \sqrt{(4-1)^2 + (6-2)^2} = 5 \text{units}$$

Question 2:

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Find the length of the line joining the point x (-4,-2) and y(3,7)

Working:

By formula, length , $L = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

Where; $x_1=-4, x_2=3, y_1=-2, y_2=7$

$L = \sqrt{(3-(4))^2+(7(-2))^2} = 11.40$units

Question 3:

If the distance between the point (16,c) and (1,1) is 17.

Find the two values of C

Working:

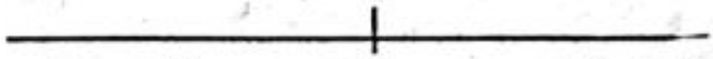
By formula, distance, $D = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

Where $x_1=16, x_2=1, y_1=C, y_2=1, D=17$

$\Rightarrow 17 = \sqrt{(1-16)^2+(1-C)^2}$, i.e. $C = -7$

But the two values of C are (-7,9),this is so because if you insert -7 o
9 in the formula, you will get 17 as the distance.

2. Midpoint Of A Straight Line:



Given, $A(x_1, y_1)$ N $B(x_2, y_2)$

then, the midpoint (N) is $N = \frac{x_2+x_1}{2}, \frac{y_2+y_1}{2}$

Question:

The midpoint of the line joining the points (6,4) and (8,2) is ____

Working:

Midpoint $N = \left[\frac{x_2+x_1}{2}, \frac{y_2+y_1}{2}\right] = \left[\frac{8+6}{2}, \frac{2+4}{2}\right] = (7,3)$

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3. *division Of A Line Segment Into A Given Ratio:*

Suppose a straight line is to be divided internally in the ratio m:n, then the point, R is given by: $R = \frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}$

Question:

Find the point which divides the line joining the point (2, -1) and (3,4) in the ratio 2:1

Working:

Given; $m=2, n=1, x_1=2, x_2=3, y_1=-1, y_2=4$

By formula, $R = \frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}$

$= \frac{2(3)+1(2)}{2+1}, \frac{2(4)+1(-1)}{2+1} = (8/3, 7/3)$

Gradient Or Slope Of A Straight Line:

By definition, gradient, $m = \frac{y_2-y_1}{x_2-x_1}$ OR $\frac{y_1-y_2}{x_1-x_2}$

$\Rightarrow y - y_1 = m(x - x_1)$

NB:

1. The gradient of a line parallel to the x-axis is 0
2. The gradient of the line parallel to the y-axis is ∞
3. If two lines are parallel to each other, then their gradients are equal. Conversely, if the gradient of two lines are equal, then they are parallel to each other.
4. The product of the gradient of two perpendicular lines is - 1

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i.e. $m_1m_2 = -1$ or $m_1 = \frac{-1}{m_2}$

Question 1:

Show that the lines $y = 2x - 7$ and $4x - 2y + 10 = 0$ are parallel to each other.

Working:

Given $y = 2x - 7$, ***this implies*** $m_1 = 2$

For $4x - 2y + 10 = 0$, we have $2y = 4x + 10$

$y = 2x + 5$, ***this implies that*** $m_2 = 2$

Since $m_1 = m_2$, ***thus***, the lines are parallel.

Question 2:

Show that the lines $y = 2x - 7$ and $y = -\frac{1}{2}x + 6$ are perpendicular to each other.

Working:

$y = 2x - 7$, $\Rightarrow m_1 = 2$. Again, $y = -\frac{1}{2}x + 6$, $\Rightarrow m_2 = -\frac{1}{2}$

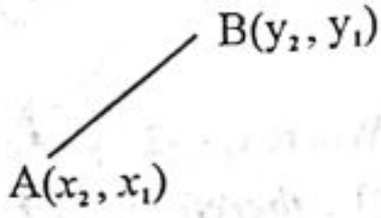
By formula, $m_1m_2 = -1$, i.e $2(-\frac{1}{2}) = -1$

Since $m_1m_2 = -1$, the line are ***perpendicular***.

Equation Of A Straight Line:

By formula, the equation of a straight line is given by $y - y_1 = m(x - x_1$

i.e.



Question 1:

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Find the equation of the line through the point (-2,5) with the slope 1/3.

Working:

Using $y - y_1 = m(x - x_1)$; Where $x_1 = -2, y_1 = 5$, $m = 1/3$

$\Rightarrow y - 5 = 1/3[x-(-2)]$; $3y - x - 17 = 0$

Question 2:

Suppose a straight line passes through the points (1,4) and (3,-2). Find its equation.

Working:

Using $m = \dfrac{y_2 - y_1}{x_2 - x_1}$. Where $x_1 = 1, x_2 = 3, y_1 = 4, y_2 = -2$

$\Rightarrow m = \dfrac{-2-4}{3-1} = \dfrac{-6}{2} = -3$

But the equation of a straight line is given by $y - y_1 = m(x - x_1)$

Substituting, we have $y - 4 = -3(x-1)$; i.e $y + 3x - 7 = 0$

Question 3:

Find the equation of the straight line through the point (-2,3) and parallel to the line $4x + 5y = 8$

Working:

Given $4x + 5y = 8$; $5y = -4x + 8$, $\Rightarrow y = \dfrac{-4x}{5} + \dfrac{8}{5}$

thus, $m = -4/5$.

But $y - y_1 = m(x - x_1)$. Where $x_1 = -2$, $y_1 = 3$

$\Rightarrow y - 3 = \dfrac{-4}{5}[x-(-2)]$; ***therefore***, $5y + 4x - 7 = 0$

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Question 4:

Find the equation of the line joining the point (1,-2) and perpendicular to the line $2x-3y=4$

Working:

Given that $2x-3y=4, \Rightarrow y=\frac{2x}{3}-\frac{4}{3}$, ***thus***, $m=\frac{2}{3}$

But $m=m_1, \Rightarrow m_1=2/3$

Remember, $m_1m_2=-1$, ***hence***, $m_2=\frac{-1}{m_1}=\frac{-1}{2/3}=\frac{-3}{2}$

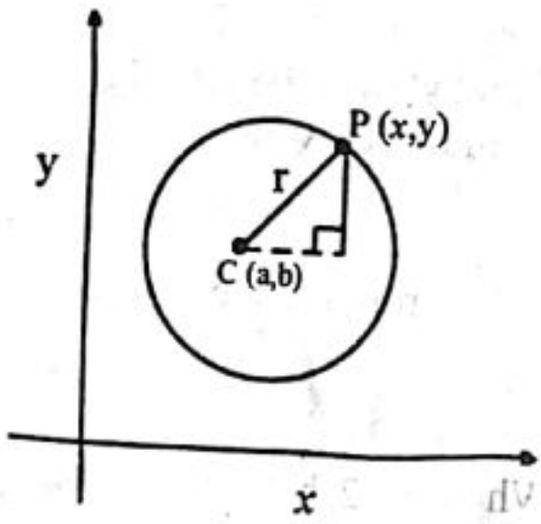
Equation of the required line is $y-y_1=m_2(x-x_1)$;

Where $x_1=1, y_2=-2$, ***thus***, $y+2=-3/2(x-1)$; $2y+3x+1=0$

THE CIRCLE:

By definition, the circle is the set of points (x, y) whose distance from fixed point C (a, b) is a constant r.

Equation Of A Circle With Centre (a,b)



Let C (a,b) be a fixed point called the center and r a fixed distance

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called the radius. Also, let P(x,y) be any point on the circle

$\therefore CP = r$ OR $\sqrt{(x-a)^2+(y-b)^2} = r$

Therefore, the equation of the circle with center (a,b) and radius r is $(x-a)^2+(y-b)^2=r^2$

This is the standard form of circle equation

Expanding, we have $x^2+y^2-2ax-2by+a^2+b^2-r^2=0$

Don't forget, if the center is at the origin (0,0) i.e a = b = 0, the equation becomes $x^2+y^2=r^2$

thus, a circle is completely determine if its center and radius are known.

The Most General Equation Of The Circle Center (-g,-f) Is Given By $x^2+y^2+2gx+2fy+c=0$

Where $r = \sqrt{g^2+f^2-c}$, i.e. $c=g^2+f^2-r^2$

Condition for Equation of a circle:

For any equation representing circle:

(1). the coefficient of x^2 and y^2 are equal with thesame positive sign

(2). there are no term in xy

(3) the coefficient of x^2 and y^2 must be one (1)

(4). the power of x and y must be two (2)

Question 1:

Find the equation of the circle through the origin with center (2,-1)

Working:

Using $(x-a)^2+(y-b)^2=r^2$; Where $a=2, b=-1$

$\Rightarrow (x-2)^2+(y+1)^2=r^2$

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Since this equation passes through the origin (0,0) we have;
$(0-2)^2+(0+1)^2=r^2$; i.e $r^2=5$. ***thus***, the required equation is
$(x-2)^2+(y+1)^2=5$, which is written as $x^2+y^2-4x+2y=0$

Question 2:

Find the equation of a circle center (3,2) and radius 4

Working:

Using $(x-a)^2+(y-b)^2=r^2$; Where $a=3, b=2, r=4$

Substituting, we have;

$(x-3)^2+(y-2)^2=4^2$. ***therefore***, $x^2+y^2-6x-4y-3=0$

Question 3:

Find the center and the length of the radius of the circle;

$x^2+y^2+4x-6y-3=0$

Working:

METHOD1:

By completing the squares in x and y, we have;

$(x+2)^2+(y-3)^2=3+4+9$

$(x+2)^2+(y-3)^2=4^2$

∴ Center = (-2,3) and $r=4$

Method 2:

Given; $x^2+y^2+4x-6y-3=0$ _____(1)

From $x^2+y^2+2gx+2fy+c=0$ _____(2)

Comparing equation (1) and (2) we have;

$2gx=4x, \Rightarrow g=2$; $2fy=-6y, \Rightarrow f=-3$

But center is $(-g,-f)=(-2,3)$

Radius, $r=\sqrt{g^2+f^2-c}$

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Where $c = -3$; $\Rightarrow r = \sqrt{2^2 + (-3)^2 - (-3)} = 4$

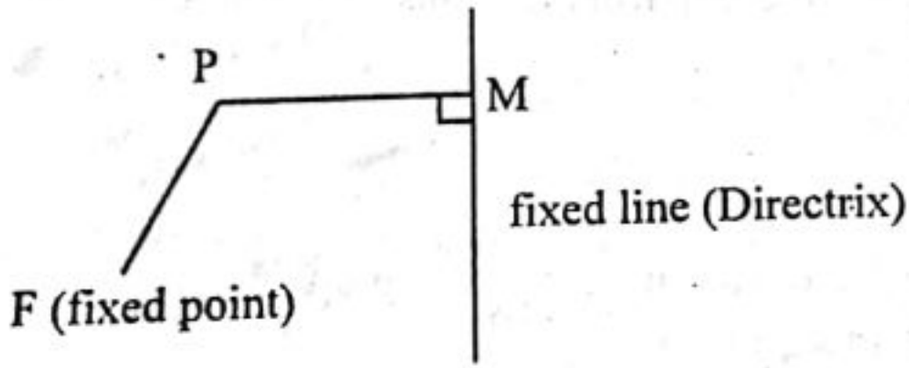
$\therefore$ Center $= (-2,3), r = 4$

Conic Section (Conic):

These are the curves one can get when one makes a straight line cut in a cone. The equations of such curve will always be second degree equations.

Note that in a conic section, there is a fixed point called the ***Focus***, a fixed line called the ***Directrix*** and a constant ratio called the ***Eccentricity*** usually denoted by 'e'

.i.e



thus, if F is a point on the locus (i.e diagram above), M is the foot of the perpendicular from P to the directrix and if

$$\frac{PF}{PM} = e$$

then, the locus of F is a conic. If $e < 1$, the conic is an *Ellipse*.

If $e = 1$, the conic is a ***Parabola***. If $e > 1$, the conic is a *Hyperbole*.

Types Of Conic Section:

(1). Parabola

(2). Ellipse (3). Hyperbole

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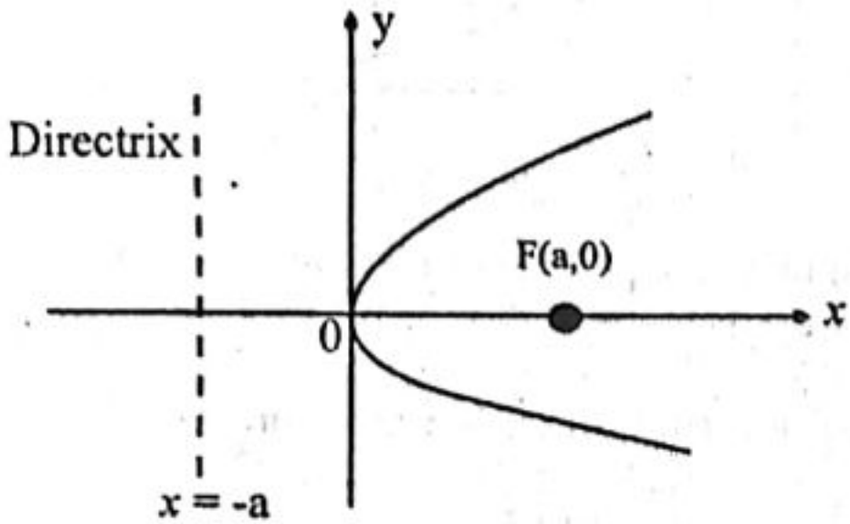
(1). Parabola (e = 1):

By definition, a parabola is a set of points in the plane which is equidistant from a fixed pont F (**called the focus**) and a fixed line L (**called the directrix**).

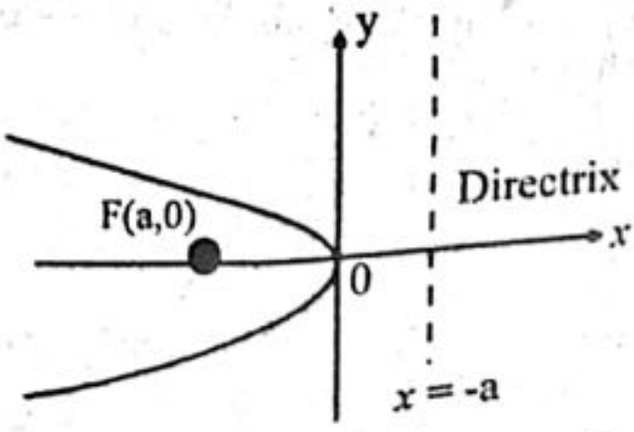
Now, let us consider the parabola with vertex at the origin and F on the Horizontal axis.

Points to note:

(1). The vertex is at the origin (0,0)
(2). The focus is along x-axis i.e F(a,0)
(3). The Directrix is the vertical line with equation as $x = -a$
(4). If $a > 0$, the parabola opens to the right and to the left if $a < 0$. i.e



(a). $y^2 = 4ax$, $a > 0$

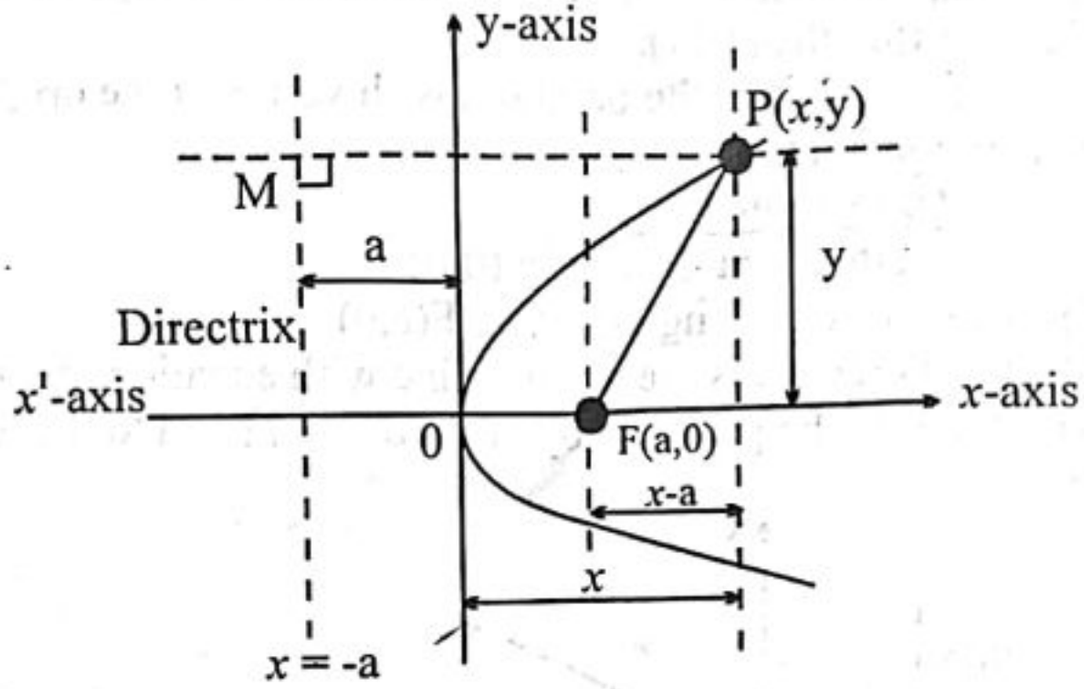


(b). $y^2 = 4ax$, $a < 0$

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Derivation of the Equation of a Parabola With F on the Horizontal Axis:

From the diagram below,



If P(x,y) is any point on the parabola (i.e diagram above), such that $\frac{PF}{PM} = e = 1$

then $(PF)^2 = (x-a)^2 + y^2$ (from Pythagoras theorem)

and $PM = x + a$

But $\frac{PF}{PM} = 1$; $\therefore (PF)^2 = (PM)^2$

$\therefore (x-a)^2 + y^2 = (x+a)^2$; i.e $x^2-2ax + a^2 + y^2 = x^2 + 2ax + a^2$

therefore, the equation of the parabola is $y^2 = 4ax$.

Note the following:

(1). Length Of Latus Rectum:

To determine the length of latus rectum, make the coefficient of y^2 to be one and the coefficient of x is the length of the latus rectum.

(2).Equation of the tangent to the Parabola at the point(x_1,y_1)

(a). For $y^2 = 4ax$, the required equation is $yy_1 = 2a(x + x_1)$

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Question 1:

Find the focus and the directrix of the parabola $3y^2 = -4x$ and obtain the length of the latus rectum.

Working:

Given that $3y^2 = -4x$

$$\Rightarrow y^2 = \frac{-4}{3}x \text{ ---------- (1)}$$

But by formula, $y^2 = 4ax$ ------- (2)

Comparing equation (1) and (2), $a = -1/3$

therefore, Focus = (a,0) = (-1/3, 0) and the equation of the directrix, $x = -a = -(-1/3) = 1/3$.

To determine the length of the latus rectum, make the coefficient of y^2 to be one and the coefficient of x is the length of the latus rectum. .i.e $3y^2 = -4x$, $\Rightarrow y^2 = \frac{-4}{3}x$

thus, the length of the latus rectum = -4/3

Alternatively, find the value of y when x = -1/3 and multiply the result by 2 .i.e $3y^2 = -4x$, $\Rightarrow y = \sqrt{\frac{-4}{3}x}$

Remember, $x = -1/3$, $\Rightarrow y = \sqrt{\frac{-4}{3} \times \frac{-1}{3}} = \frac{2}{3}$

Hence, the length of the latus rectum = $2 \times \frac{2}{3} = 4/3$

Question 2:

Find the focus and the directrix of the parabola $y^2 = -6x$ and sketch the graph

Working:

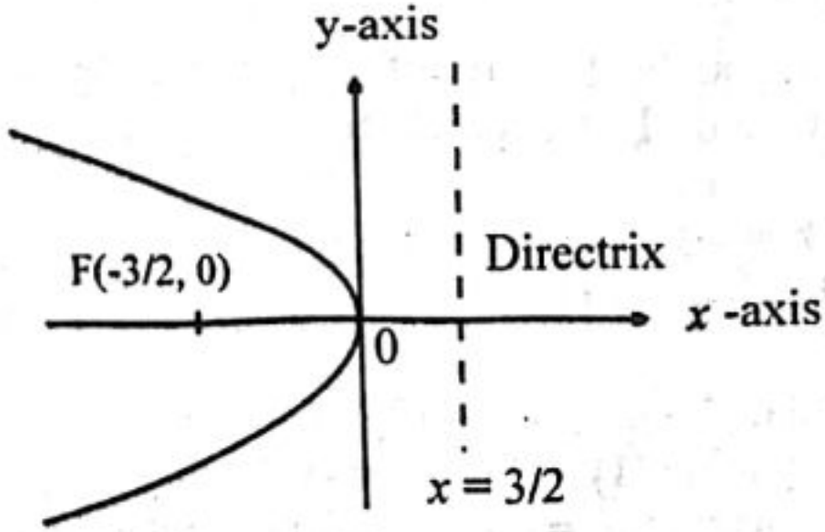
Given $y^2 = -6x$ ---------- (1)

But $y^2 = 4ax$ ---------- (2)

Compare equation (1) and (2), $a = -3/2$; .: Focus = (a,0) = (-3/2, 0) and the equation of the directrix, $x = -a = -(-1/3) = 1/3$

The graph is shown below:

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Question 3:

Find the equation of the parabola with focus (4,0) and directrix, $x = -6$

Working:

Let P(x, y) be any point on the parabola, then P is equidistant from the focus and the directrix. thus, PF = PM

i.e $(x-4)^2 + (y-0)^2 = (x+6)^2$, *thus*, $y^2 = 20(x+1)$

NB: the length of the latus rectum is 20

Question 4:

Find the equation of the parabola (4,3) and directrix, $x = 2$.

Working:

Hint: Use $(x-4)^2 + (y-3)^2 = (x-2)^2$

Now, let us consider the one with vertex at the origin and F on the vertical line

The graph of the equation $x^2 = 4ay$ is a parabola with the following properties:

(i). Vertex at the origin (0, 0)
(ii). Focus along y-axis F(a, 0)
(iii). Directrix is the horizontal line y = -a
(iv). The graph opens upward if a > 0 and downward if a < 0

Question 1:

Find the focus and directrix of the parabola $y = -x^2$, sketch the

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graph.

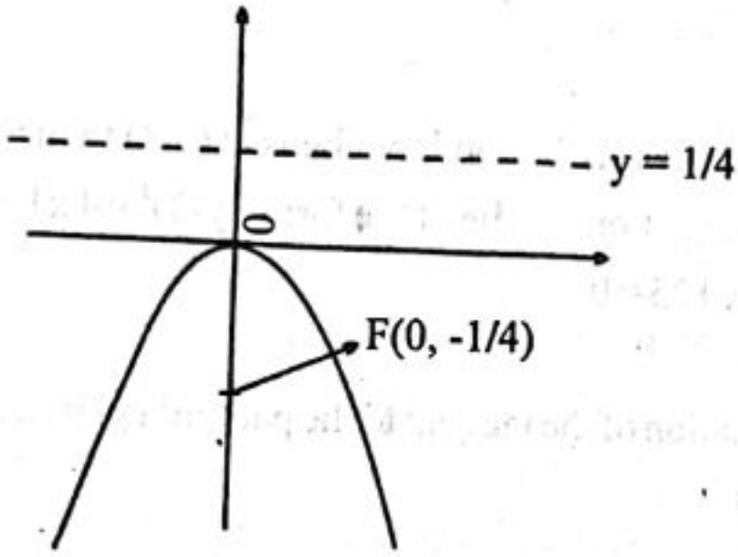
Working:

$x^2 = -y = -1(y)$ ------- (1)

But $x^2 = 4ay$ --------- (2)

Comparing (1) and (2), we have a = -1/4. ***thus***, the directrix is y = 1/4, focus is F(0, -1/4).

.i.e.



Question 2:

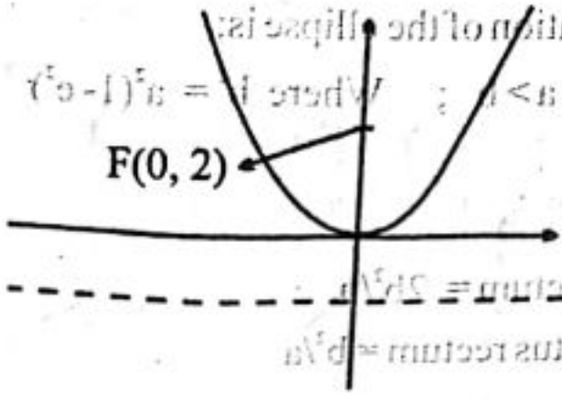
Find the equation of the parabola with vertex at (0, 0) and focus F(0, 2). Sketch the graph also.

Working:

Since focus is F(0,2) , a = 2, thus directrix is y = -2

therefore, the equation is $x^2 = 4(2)y = 8y$.

Since a > 0, the parabola opens upward. i.e



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Shifted Parabolas:

Use the Equations below:

(1). $(y-k)^2 = 4a(x-h)$

(2). $(x-k)^2 = 4a(y-h)$

Question 1:

Derive the equation of the parabola with its vertex at (4,3) and its focus at (5,3)

Working:

Since the vertex is at (4,3) and the focus at (5,3) then $a = 5-4 = 1$.

the required equation will be of the form $(y-3)^2 = 4 \times 1 (x-4)$

i.e $y^2 - 6y - 4x + 25 = 0$

Question 2:

Find the equation of the tangent to the parabola $y^2 = 5x$ at the point (-2,4)

Working:

For the parabola, $y^2 = 8x$ we have $a = 2$, $(x_1, y_1) = (-2,4)$

But $yy_1 = 2a(x+x_1)$

$\Rightarrow 4y = 2(2)(x-2)$; i.e $4y - 4x = -8$

ELLIPSE (e<1):

The ellipse is a conic section whose eccentricity is less than 1.

the standard form of equation of the ellipse is;

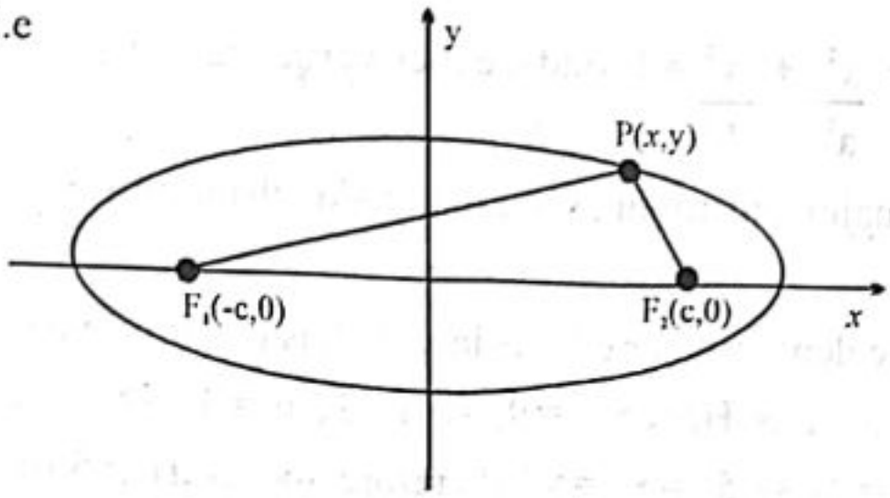
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \text{ With } a > b \text{ ; Where } b^2 = a^2(1-e^2)$$

NB:

1. Length of the latus rectum $= 2b^2/a$
2. Length of the semi latus rectum $= b^2/a$

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i.e



Question 1:

Find the eccentricity e and the length of the latus rectum of the ellipse $2x^2+3y^2=5$

Working:

The equation can be put in the standard form as;

$$\frac{x^2}{5/2}+\frac{y^2}{5/3}=1 \quad ; \text{ So, } a^2=5/2,\ b^2=5/3$$

thus, for e, we use $b^2=a^2(1-e^2)$

i.e. $5/3=5/2\,(1-e^2)$; $\therefore e=1/\sqrt{3}=\sqrt{3}/3$

The length of latus rectum $=\frac{2b^2}{a}=\frac{2(5/3)}{\sqrt{5/2}}=\frac{10}{3\sqrt{5/2}}$

Question 2:

Find the equation of the ellipse with vertex $(\pm 4, 0)$ and foci $(\pm 2, 0)$.

Working:

Given, $a=4, c=2$. From $b^2=a^2-c^2$; $\Rightarrow b^2=4^2-2^2=16-4=12$

thus, the equation becomes $\frac{x^2}{16}+\frac{y^2}{12}=1$ $\left[\text{i.e. from } \frac{x^2}{a^2}+\frac{y^2}{b^2}=1\right]$

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Question 3:

For the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, find the foci, vertices and the length of the major and minor axis. Finally, sketch the ellipse.

Working:

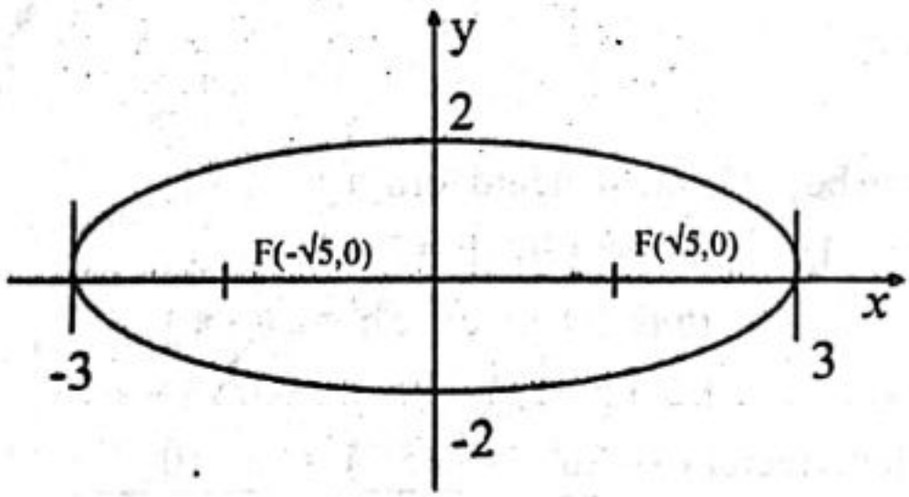
Notice that the denominator of x^2 is larger, thus the ellipse has a horizontal major axis. Hence, $a^2=9, \Rightarrow a=3$; $b^2=4, \Rightarrow b=2$.

But $c^2=a^2-b^2=9-4=5, \Rightarrow c=\sqrt{5}$. therefore, foci is at $(\pm\sqrt{5},0)$

Vertices are at $(\pm3,0)$. Length of major axis $= 2a = 2x3 = 6$,

Length of minor axis $=2b=2x2=4$.

i.e



Note: By formula, eccentricity $e = \frac{c}{a}$ Where $c = \sqrt{a^2-b^2}$

The eccentricity of every ellipse satisfies $0<e<1$

Example:

Find the equation of the ellipse with foci at $(0, \pm8)$ and eccentricity $=4/5$

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Working:

By formula, $e = \frac{c}{a}$; $\Rightarrow \frac{4}{5} = \frac{8}{a}$, $\Rightarrow a = 10$. ***thus***, $c^2 = a^2 - b^2$

i.e $8^2 = 10^2 - b^2$

therefore, $b = 6$

Hence the equation becomes $\frac{x^2}{36} + \frac{y^2}{100} = 1$

Shifted Ellipse:

If we shift $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, so that its centre is at the point(h,k)

instead of the origin, then the equation becomes

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

Question 1:

Analyze $\frac{(x+1)^2}{4} + \frac{(y-2)^2}{9} = 1$

Working:

The center of this ellipse is (-1,2). This is obtained by shifting $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, 1 unit to the left and 2 units upwards.

The ellipse has a vertical major axis with $a = 3, b = 2$

Question 2:

Analyze the equation $9x^2 + 4y^2 + 36x - 8y + 4 = 0$

Working:

By Completing the Squares i.e. $9x^2 + 4y^2 + 36x - 8y = -4$

$$9(x^2 + 4x) + 4(y^2 - 2y) = -4$$

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$9(x^2+4x+4)+4(y^2-2y+1)=-4+36+4$

$9(x+2)^2+4(y-1)^2=36$

Dividing both sides by 36

$\therefore \frac{(x+2)^2}{2^2} + \frac{(y-1)^2}{3^2} = 1$

This is the equation of an ellipse with center at (-2,1) intercepts at $(0, \pm 3)$ on the y – axis, and $(\pm 2, 0)$ on the x-axis.

To locate foci, use the relation $c = \sqrt{a^2-b^2} = \sqrt{9-4} = 5$

Where $a=3$ and $b=2$

Then the eccentricity e is $e=c/a=\sqrt{3}/3$

Therefore, the equation of the directrix are $y=\pm a/e$

i.e. $y=\pm \frac{3}{\sqrt{5}/3}$ or $\frac{9\sqrt{5}}{5}$

The length of the latus rectum of the ellipse is $\frac{2b^2}{a} = \frac{2(2)^2}{3} = \frac{8}{3}$

Further Examples:

Question:

Find (I) the eccentricities (ii) the coordinates of the foci (iii) the equations of the directrix of the ellipse $\frac{x^2}{100} + \frac{y^2}{64} = 1$

Working:

Given; $\frac{x^2}{100} + \frac{y^2}{64} = 1$ _____(1)

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By formula, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ____(2)

Compare equation (1) and (2)

$a^2 = 100, \Rightarrow a = \pm 10$, $b^2 = 64, \Rightarrow b = \pm 8$

(i). $b^2 = a^2(1-e^2)$; $64 = 100(1-e^2)$, $\therefore e \pm 3/5$

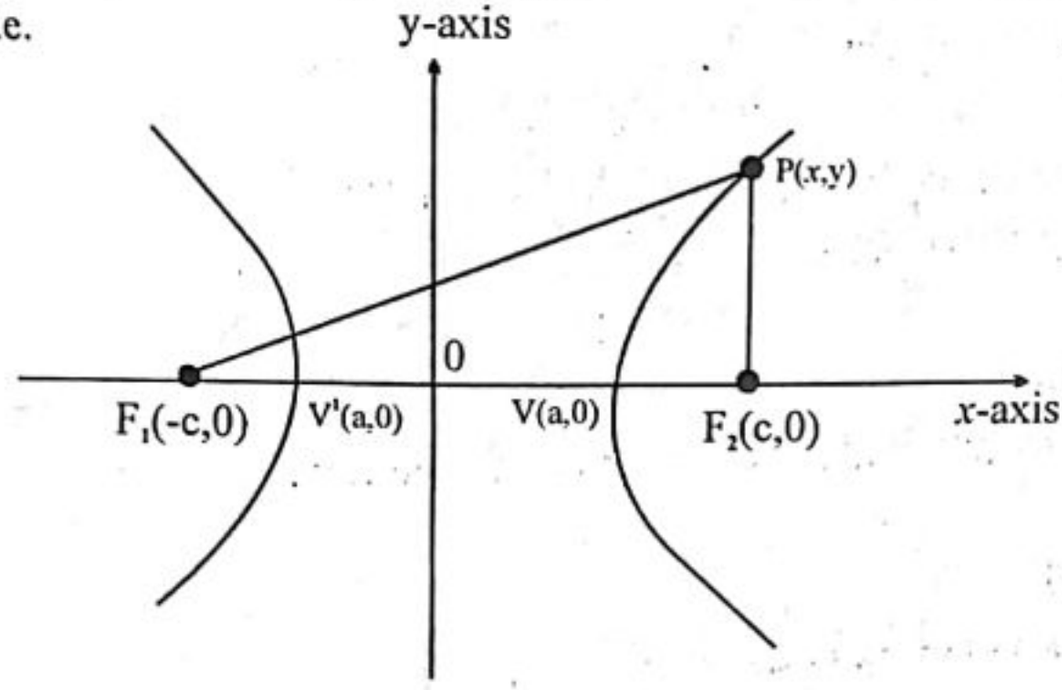
(ii). the coordinates of the foci are $(\pm a, 0) = [\pm 10(3/5), 0] = (\pm 6, 0)$.

iii. the equation of the directrix are $x = \pm ae = \pm 10(3/5) = \pm 30/5 = \pm 6$

Hyperbola (e>1):

The hyperbola is a conic section whose e> 1 given by $b^2 = a^2(e^2 - 1)$.

i.e.



The standard equation of the Hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$,

with center at the origin. The corresponding points V(a,0) and $v'(-a,0)$ are the vertices and the segment VV' is known as the

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transverse axis. The foci are the point $(\pm a/e, 0)$ and the directrices to the line is $x = \pm a/e$

Question 1:

Find the equation of the hyperbola with its center at the origin, transverse axis on the x-axis and which passes through points (3,1) and (9,5)

Working:

Since the transverse axis is on the x-axis, use

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad ; \quad \therefore \quad \frac{9}{a^2} - \frac{1}{b^2} = 1 \text{ and } \frac{81}{a^2} - \frac{25}{b^2} = 1$$

With points (3,1) and (9,5)on the hyperbola, substitute for the x and y coordinates that is $9b^2 - a^2 = a^2b^2$ _____(1)

$81b^2 - 25a^2 = a^2b^2$ _____2)

Solving (1) and (2) simultaneously, $a^2 = 6$ and $b^2 = 2$

The equation now becomes $\frac{x^2}{6} - \frac{y^2}{2} = 1$ or $x^2 - 3y^2 = 6$

Question 2:

For the hyperbola $9x2 - 16y2 = 144$, find the vertices, foci and asymtotes.

Working:

Divide each term by 144 to get $\frac{x^2}{16} - \frac{y^2}{9} = 1$

The hyperbola has a horizontal axis since $\frac{x^2}{16} > 0$

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$a^2 = 16$, $\Rightarrow a = 4$; $b = 3$, $c = \sqrt{a^2 + b^2} = 5$, $\therefore$ vertices $= (\pm 4, 0)$,

Foci at $(\pm 5, 0)$; asumptotes $= \pm \frac{b}{a} x = \pm \frac{3x}{4}$

Question 3:

Find the equation of the hyperbola with vertices at $(\pm 3, 0)$ and foci a $(\pm 4, 0)$.

Working:

The equation is, $\frac{x^2}{3^2} - \frac{y^2}{b^2} = 1$

But $c^2 = a^2 + b^2$ .i.e $4^2 = 3^2 + b^2$, $\Rightarrow b^2 = 4^2 - 3^2 = 7$

therefore, the equation is $\frac{x^2}{9} - \frac{y^2}{7} = 1$

Shifted Hyperbolas:

When the center shift from the origin to (h,k), we may have equations of the form:

(a). $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$ or (b). $\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$

Question 1:

Analyze and sketch the equation $9x^2 - 16y^2 - 18x - 64y - 199 = 0$

Working:

Complete the squares and write the equation in the form

$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$

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$\therefore 9(x^2-2+1)-16(y^2+4y+4)=199-64-9$

i.e $9(x-1)-16(y+2)^2=144$

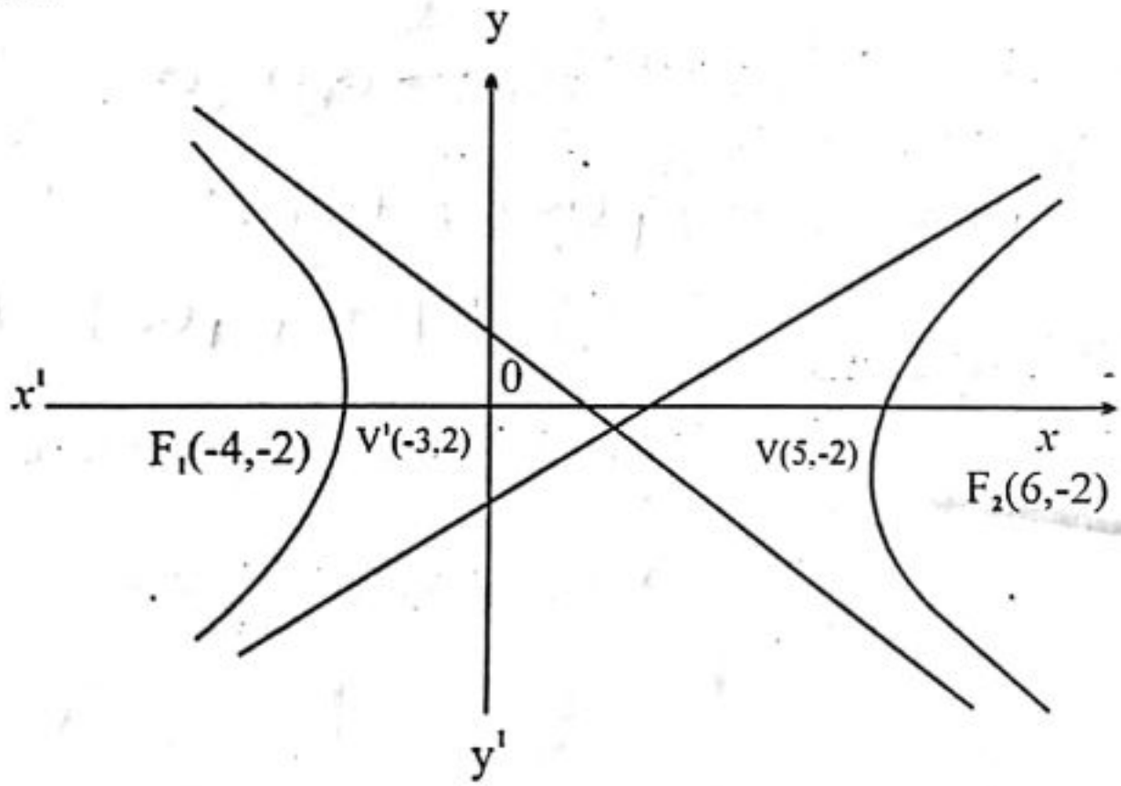
$\therefore \frac{(x-1)^2}{16} - \frac{(y+2)^2}{9}=1$

thus, $a=4,\ b=3,\ \therefore c=\sqrt{16+9}=5$

:., the center is at (1,-2), the vertices are (5, -2) and (-3,2) while the foci are (6,-2) and (-4,-2) on $x-$axis.

The equations of the directrices are $x=\pm a/e=\pm 4(4/5)=\pm 16/5$

i.e.



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ABOUT THE AUTHOR

Titigwaye Mathew . O. (A.K.A Mr. Ohms), Obtained his B.Sc in Applied Microbiology and Brewing, 2005 (2nd Class) from Nnamdi Azikiwe University, Awka. He is an itinerant tutor. He lectures undergraduates students tutorials of various Universities

FOR CONTACT

Mr. Ohms Bookshop
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